CMPS 4200 HCI: AI-ENHANCED UI/UX · PHASE 3

Farshore

Usability testing and iteration of an AI-rival arcade journey through Homer's *Odyssey*

Student	Bibek Pathak (solo project)
Date	November 5, 2026
Figma file	**Figma view link (to be added)**
Repository	**GitHub repository link (to be added)**
Raw data	`Farshore-Phase-3-Test-Data.xlsx` (submitted separately)
AI in the product	Qwen3-4B-Instruct plans each rival and writes live commentary; Kokoro-82M speaks it

Contents

1. Screen designs: style basics, AI states and user control, screens
2. Test script
3. Test report
4. List of changes
5. AI tool use statement
6. Team contribution statement

1. Screen designs

High-fidelity screens of the tested prototype, captured from the running game and annotated. The same frames are in the Figma file (**link to be added**). Numbered markers match the notes beside each screen: gold = interface, cyan = AI behavior or AI state. Failure states were produced with the mock AI service (Section 1.2), and the interface shown is the real one.

Flows: F1 Start, F2 Choose a trial, F3 Prepare the rival (the AI planning states), F4 Play, F5 Understand the rival (explainable AI), F6 Settings, F7 Account. Personas are the Phase 1 sketches: Maya (gamer, fairness), Daniel (reader, limited technical comfort), Rosa (AI skeptic). What changed from the Phase 2 wireframes, and why, is in Section 4, Part A.

1.1 Style basics (design system)

Decided before the screens and reused on every one. The values below are read from `src/styles/tokens.css`, the single file where the app defines its colours, type and spacing, when this document is built. Every stylesheet reads these variables.

Colour palette

Background `--fs-color-bg`
#071523
Page and stage background

Surface `--fs-color-surface`
#0b2033
Panels and cards

Text `--fs-color-text`
#f6f0e3 · 16.2:1 on background
Primary text

Muted text `--fs-color-text-muted`
#93afc0 · 8.0:1 on background
Secondary lines and status text

Accent `--fs-color-accent`
#e9b85c · 10.1:1 on background
The one accent: buttons, focus, interactive chrome

AI `--fs-color-ai`
#32d4f5 · 10.4:1 on background
Only for AI state and AI data, never for hazards

Error `--fs-color-error`
#f4604f · 5.8:1 on background
Errors and real threats

Success `--fs-color-success`
#9fe6b4 · 12.7:1 on background
Confirmation and successful escapes

One accent, gold, for everything a player can press. Cyan is reserved for the AI, so "the rival is thinking" is never confused with "you are in danger" (red). Contrast ratios use the WCAG formula against the background. A few extra navy, gold and bronze tints are used only for depth and highlights.

Type: four families, five sizes

Family	Stack	Specimen	Use
Display `--fs-font-display`	Georgia, "Palatino Linotype", serif	Eight Trials. One Journey Home.	Story, headings, the rival panel
UI `--fs-font-ui`	Arial, Helvetica, sans-serif	Eight Trials. One Journey Home.	Small interface details
Pixel `--fs-font-pixel`	Pixel, monospace	Eight Trials. One Journey Home.	Arcade labels and eyebrows

Family	Stack	Specimen	Use
Mono `--fs-font-mono`	ui-monospace, SFMono-Regular, Consolas, monospace	Eight Trials. One Journey Home.	Inside the AI inspector only

Size	Value	Specimen	Use
XS `--fs-text-xs`	12px	The rival is studying you	Captions, status lines, labels
S `--fs-text-sm`	14px	The rival is studying you	Body text and buttons
M `--fs-text-md`	16px	The rival is studying you	Panel titles, quotes, lead text
L `--fs-text-lg`	24px	The rival is studying you	Section headings
XL `--fs-text-xl`	clamp(40px, 6vw, 72px)	The rival is studying you	Screen titles (scales with the window)

Spacing scale

`--fs-space-1`	4px
`--fs-space-2`	8px
`--fs-space-3`	12px
`--fs-space-4`	16px
`--fs-space-5`	24px
`--fs-space-6`	32px
`--fs-space-7`	48px

In code: `tokens.css` defines the values, and the older colour names in `farshore-stage.css` are aliases of them. The AI rival panel, in all five states, uses only tokens. In the older stylesheets, every value equal to a token was replaced with that token. Off-scale values remain in in-game HUD labels and some menu rules, and are listed for Phase 4 (`docs/UI_DESIGN.md`).

1.2 AI components: five states and user control

The main AI feature is the rival's strategy. Before each round, the model reads the player's recent attempts and chooses how the mythic rival will play. It appears in the briefing's rival panel (flow F3) and is designed in all five states. The explainable AI Insights panel in Cyclops' Cave has its own empty state (D23).

State	What the player sees	Frames
1. Empty / initial	The briefing opens before any strategy exists. Preparation starts at once, so the panel opens in the loading design at 0 s rather than as an empty box. With no model connected, the initial state is "No AI rival connected", with three ways to connect and Check again. In Cyclops' Cave, AI Insights reads "Observing. No committed action." until the rival's first decision.	D06, D10, D23
2. Loading	"[Rival] is studying you", an elapsed-seconds counter and a cyan scanning bar, which does not pretend to know the remaining time. Matched to measured latency: the Phase 2 spike measured 7.3-11.6 s (mean 8.3 s) per strategy, and a CPU laptop with Qwen3-4B takes about 15 s. The mock's slow mode (12 s) tests this. Cancel, or Keep the previous strategy if there is one.	D06, D25
3. Success	The strategy name, "Strategy chosen by [model]", a plain description, the rival's taunt (with Hear it), and Ask for another strategy. Start is enabled.	D07, D13
4. Low confidence	"The rival's plan was set aside". The model's strategy did not pass the game's checks, so it was not used and nothing changed. The main action is Ask for another strategy. Keep the previous strategy appears if one exists. Amber, not red: nothing is broken.	D22
5. Error, with recovery	"The rival is not ready", with a plain message and no codes. One clear action, Prepare the rival again, plus "Nothing about your voyage changes". Keep the previous strategy appears if one exists. Task 4 of the test leads participants here.	D09

User control over AI results

Control	How it works	Frames
Regenerate	"Ask for another strategy" on a ready strategy, and as the main action in the low-confidence state. The model chooses again.	D07, D24
Undo	After asking again, "Undo: back to [previous strategy]" restores it, and pressing Undo again swaps back. While a new strategy is loading, or if it fails or is set aside, "Keep the previous strategy" returns to the old one.	D24, D25, D22
Reject	The game rejects a strategy that fails its checks before it can be used (low-confidence state). Players can reject the AI commentary by switching commentary or voice off in Settings, or with the Voice button on the menu.	D22, D18
Cancel	"Cancel preparation" stops a request in progress. The pilot found that the briefing then starts a new request at once. This is noted for the sessions.	D06
Edit	Not offered by design: the strategy is the opponent's move. Changing the difficulty asks the model for a new strategy at the new level.	D08

Mock AI service

All AI calls (rival status, strategy and live commentary) go through one module, `server/lib/model-api.ts`. It asks `server/lib/mock-ai.ts` first. The mock answers in the modes success, slow, uncertain, error and offline, plus "-once" modes that fail once and then recover. A moderator switches modes from a second browser tab at `/api/ai-mock?mode=...`, or for the whole server with `FARSHORE_AI_MOCK`. Farshore's live model is already connected (change A2), so in Phase 4 the mock stays as the test fixture for these states instead of being replaced. Frames marked "mock" were captured in the real interface with the mock answering, and name the model as "mock-ai (test mode)".

1.3 Desktop screens (1440 × 900)

D01 Splash

F1 Start · All players



1. One call to action: press Enter or click to start. Keyboard, mouse and touch are all named.
2. Sign in is optional. Guests save on this device, as the top bar says.

D02 Title menu

F1 Start · Daniel, Maya

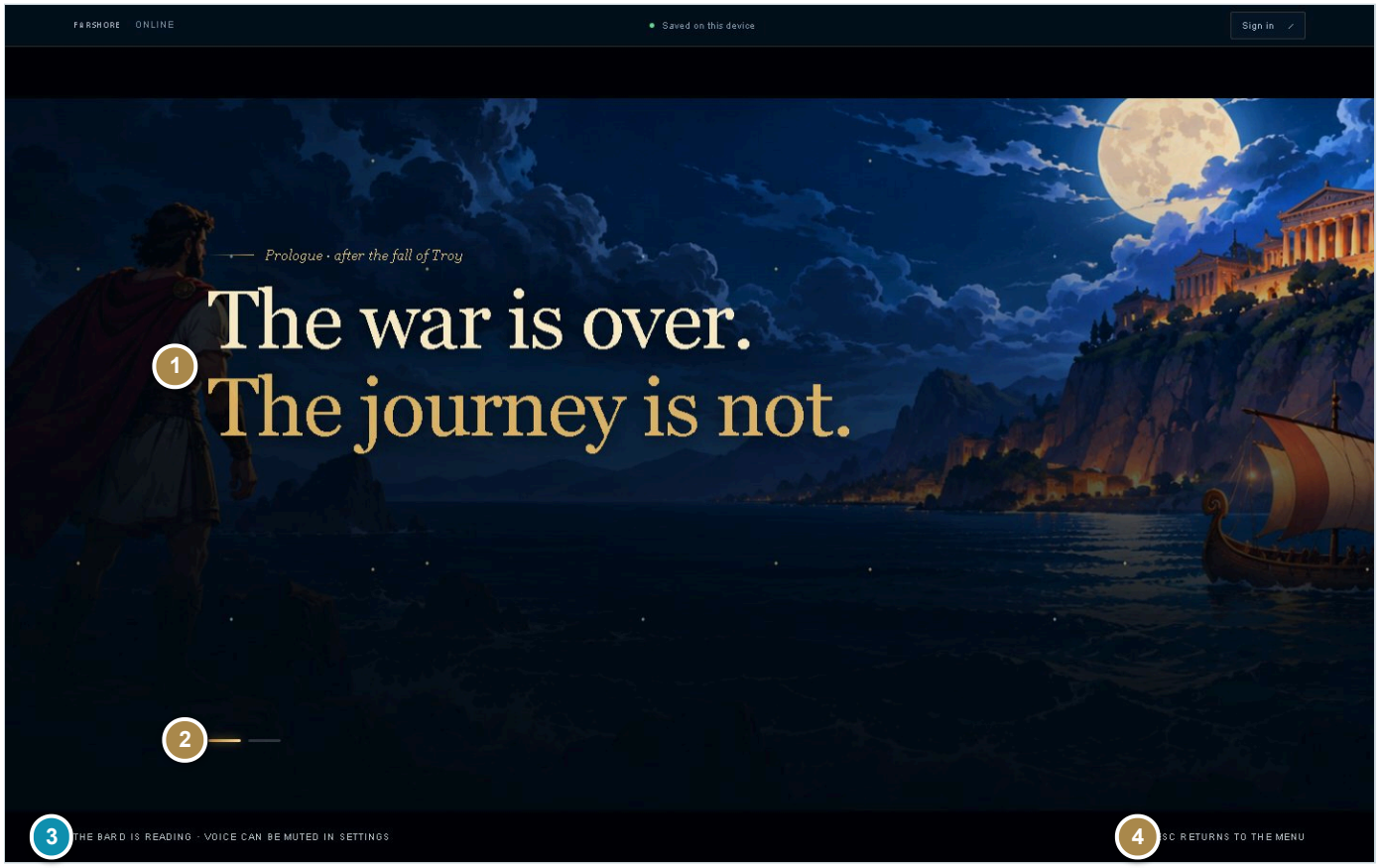


AI state: Rival status chip: checking, online (with model name) or offline (with Connect)

1. New game is the highlighted primary action. Every entry has a one-line explanation (T1).
2. Continue explains why it is unavailable: "No voyage saved yet".
3. The AI rival's status is visible before play, so a missing model is never a surprise mid-game (T4).
4. Voice and sound toggles on the first screen, without opening Settings.
5. AI Lab: watch the rival's decision pipeline without affecting the campaign.

D03 Prologue

F1 Start · Daniel

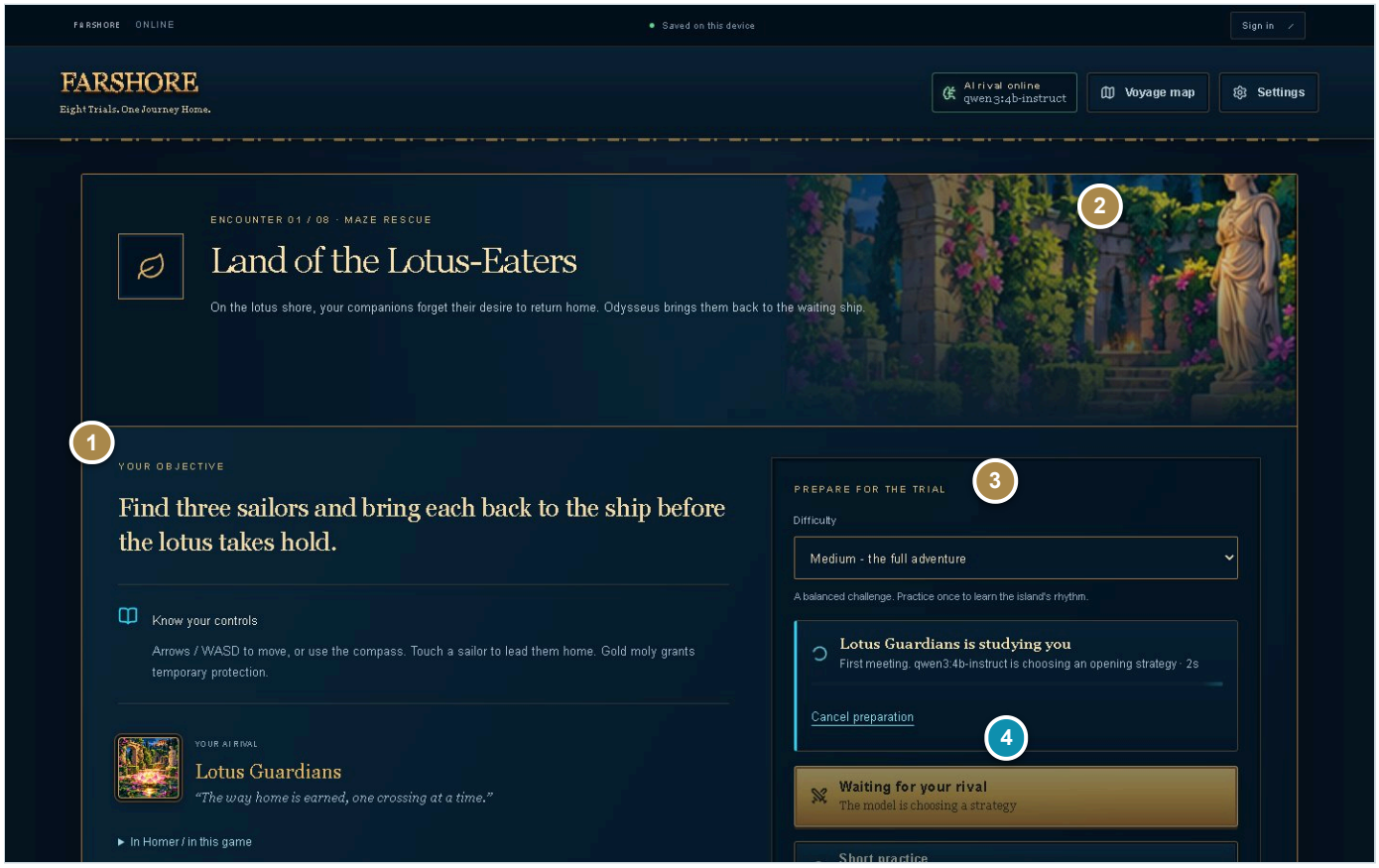


AI state: Bard voice (Kokoro-82M) reads each card aloud

1. Two short cards set the story. Large type, one idea per card.
2. Progress dots show how many cards remain.
3. States that the bard is reading and where to mute the voice.
4. Escape returns to the menu. The prologue can also be skipped to the voyage map.

D04 Trial selection

F2 Choose a trial · Maya, Daniel

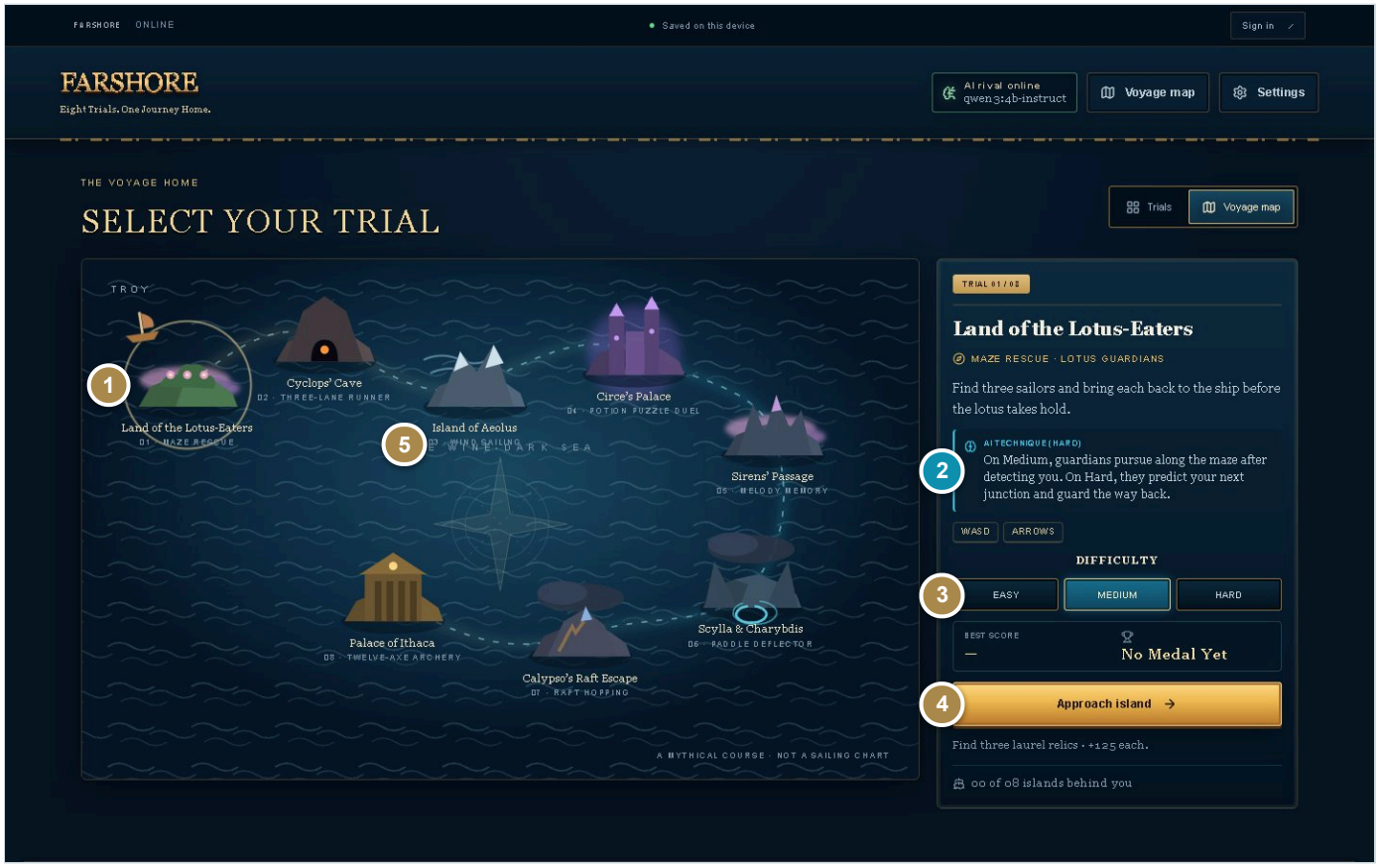


AI state: Per-trial note on the AI technique the rival uses at each difficulty

1. Eight numbered cards in voyage order. The selected trial has a gold outline.
2. Switch between the trial gallery and the illustrated voyage chart.
3. The side briefing previews the objective before the player commits.
4. Explains how the rival behaves on Medium and Hard, in plain words (transparency for Rosa).

D05 Voyage chart

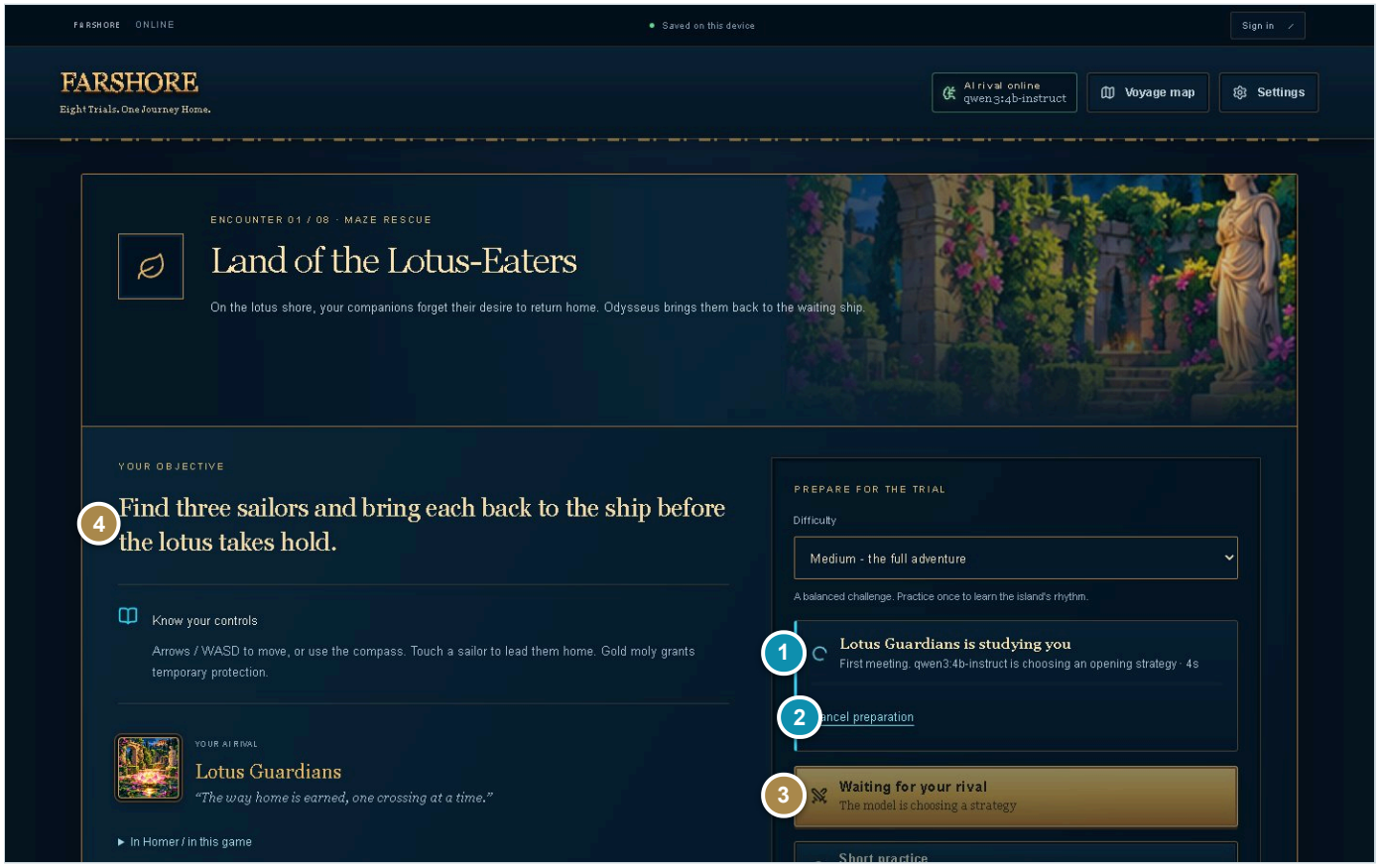
F2 Choose a trial · Daniel



1. The current island is ringed, and the dashed route shows the voyage order from Troy to Ithaca.
2. The same AI technique note as the gallery, beside the chart.
3. Difficulty can also be chosen here.
4. Approach island opens the briefing.
5. Pilot: some island labels overlap the "wine-dark sea" caption.
6. Locked, open and cleared islands differ by icon and label, not colour alone.

D06 Briefing: the rival is preparing (states 1-2 of 5: initial and loading)

F3 Prepare the rival (A1 loading) · Rosa, Maya

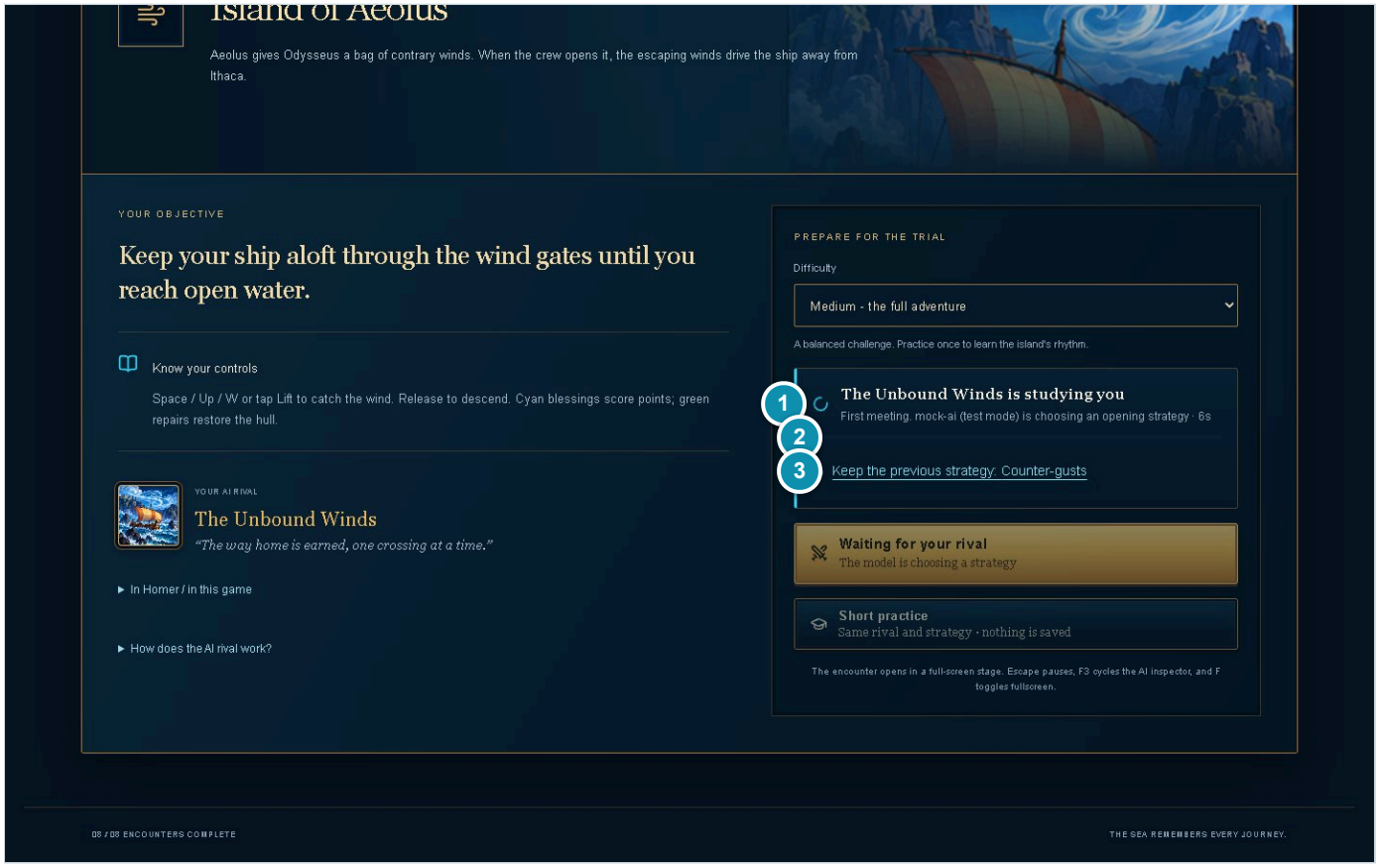


AI state: A1 loading: the model reads recent attempts and chooses a strategy (about 15 s on a CPU laptop)

1. Says what the model is doing ("is studying you"), names the model, says whether it is a first meeting, and counts the elapsed seconds.
2. Cancel preparation: the player is never stuck waiting.
3. Start says "Waiting for your rival" and is disabled until a plan is ready.
4. The objective and controls can be read while the model works, so the wait is useful.

D25 Briefing: slow response while asking again (state 2 of 5: loading)

F3 Prepare the rival (A1 loading) · Rosa, Daniel

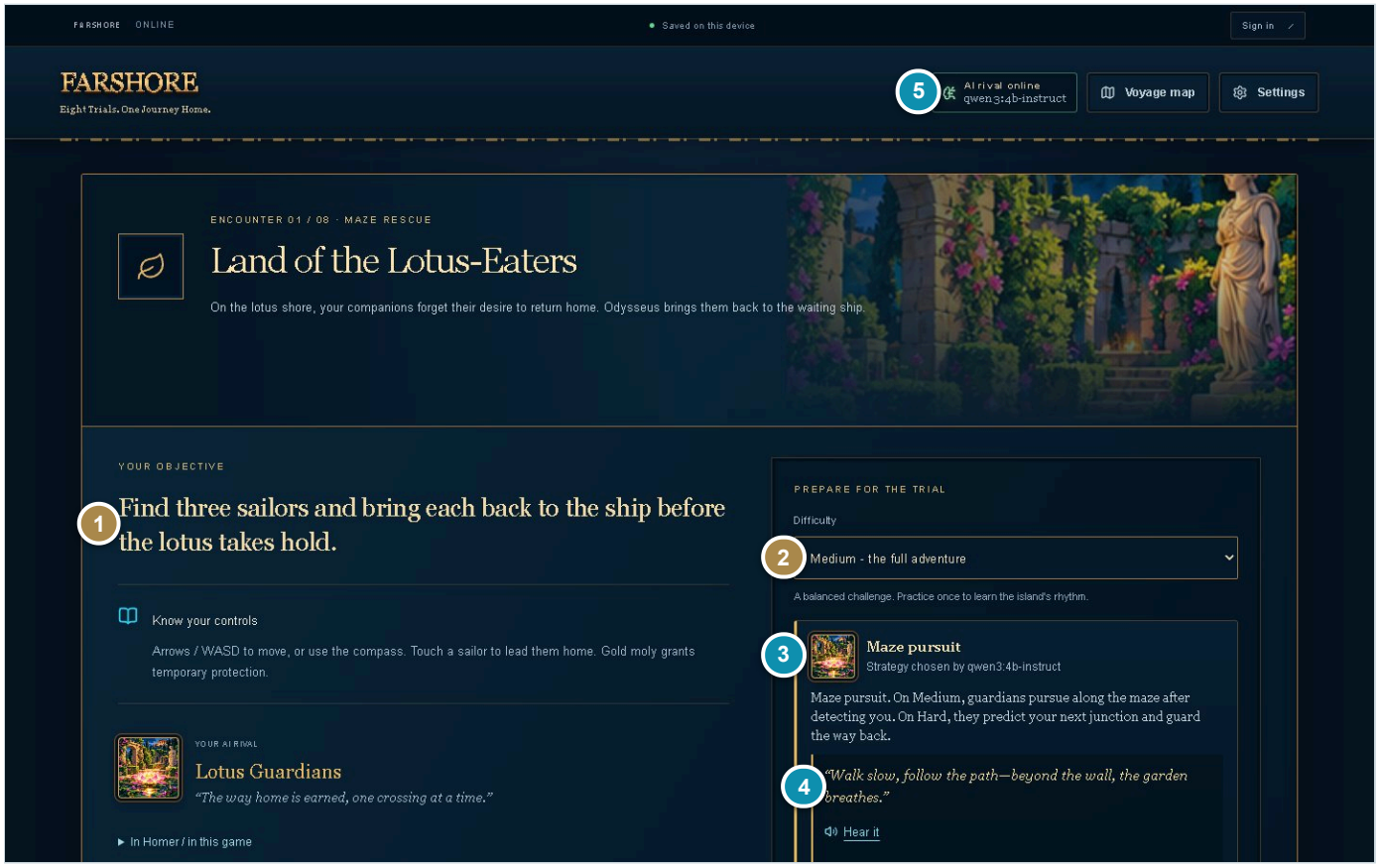


AI state: Loading during a slow response. Mock AI service in mode slow (12 s, just above the slowest strategy measured in the Phase 2 spike, 11.6 s)

1. Elapsed seconds instead of a fake percentage: the model's time varies, so the bar only shows activity.
2. Cyan scanning bar: cyan is used only for AI activity.
3. While a new strategy loads, the previous one is one click away (undo), so a slow model never blocks play.
4. Captured in the real interface with the mock AI service answering, so the model is named "mock-ai (test mode)".

D07 Briefing: strategy ready (state 3 of 5: success)

F3 Prepare the rival (A1 success) · Rosa, Maya

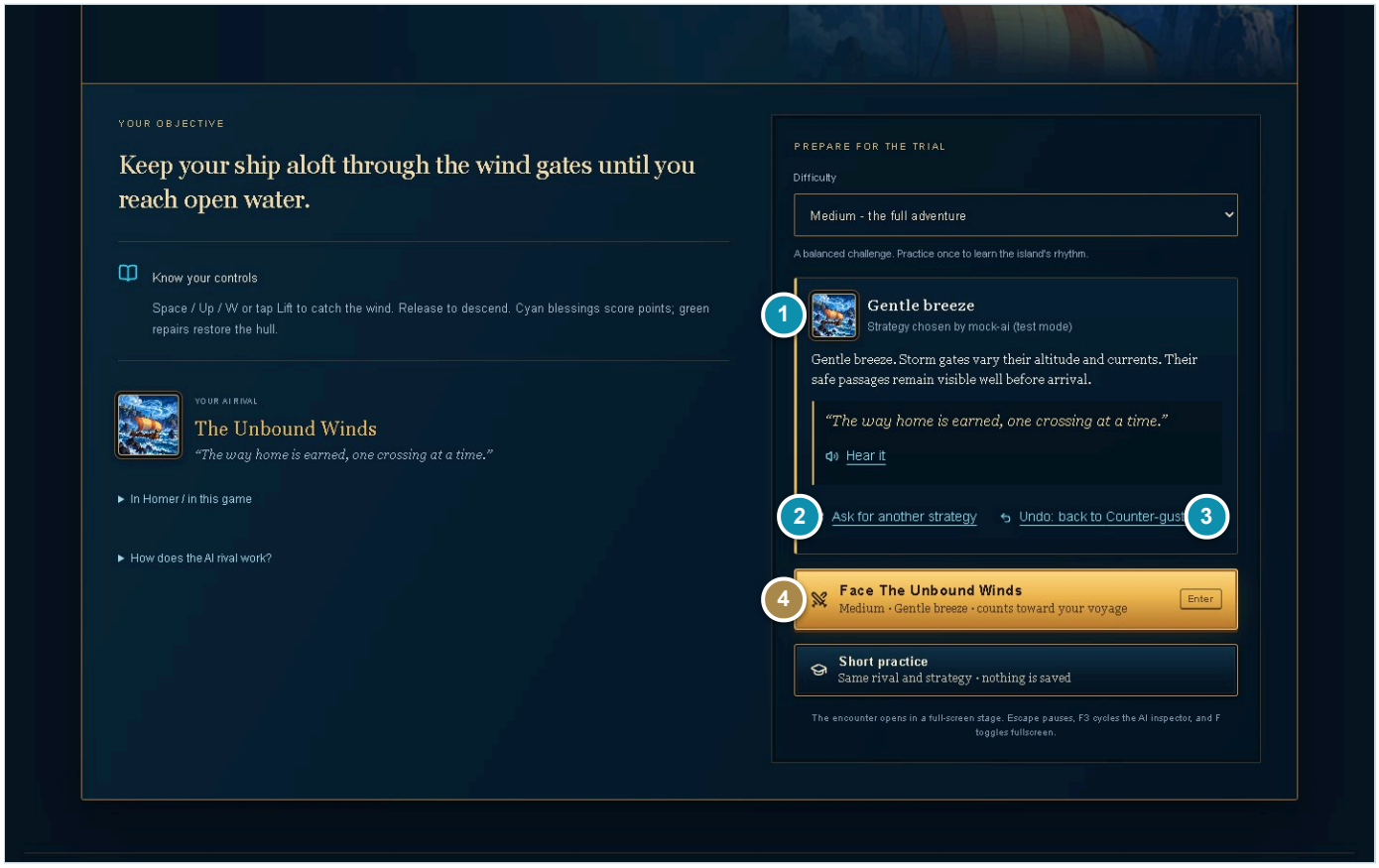


AI state: A1 success: named strategy, who chose it, description and taunt

1. The objective comes first, in plain words.
2. Difficulty with a plain-language note. Pilot: it defaults to Medium, which new players may not notice.
3. Strategy name and "Strategy chosen by qwen3:4b-instruct": the AI's decision is labelled as the AI's (T2).
4. Model-written taunt, with Hear it to play it in the rival's voice.
5. The rival status chip stays visible on every menu screen.

D24 Briefing: another strategy, with undo (user control)

F3 Prepare the rival (A1 success, regenerate and undo) · Maya, Rosa

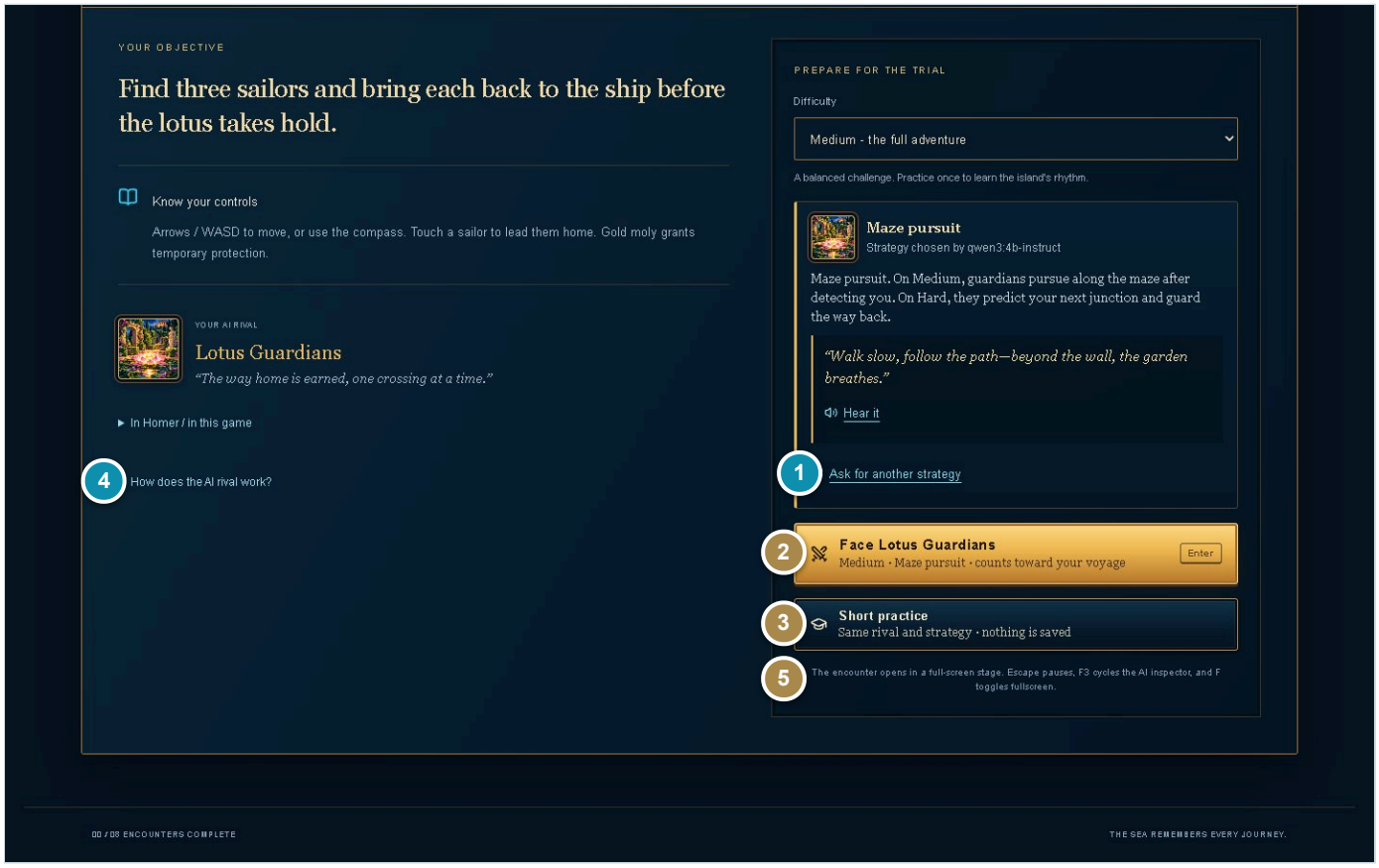


AI state: User control: the player asked for another strategy (Counter-gusts became Gentle breeze) and can undo. Mock AI service in mode success

1. The new strategy, and who chose it: "Strategy chosen by" the model.
2. Regenerate: Ask for another strategy.
3. Undo: back to the strategy the player had before. Pressing it again swaps back.
4. The start button repeats the chosen strategy, so the player knows what they are accepting.
5. Edit is not offered by design: the strategy is the opponent's move. Changing difficulty asks for a new one.

D08 Briefing: controls and start

F3 Prepare the rival (A1 success) · Daniel, Rosa

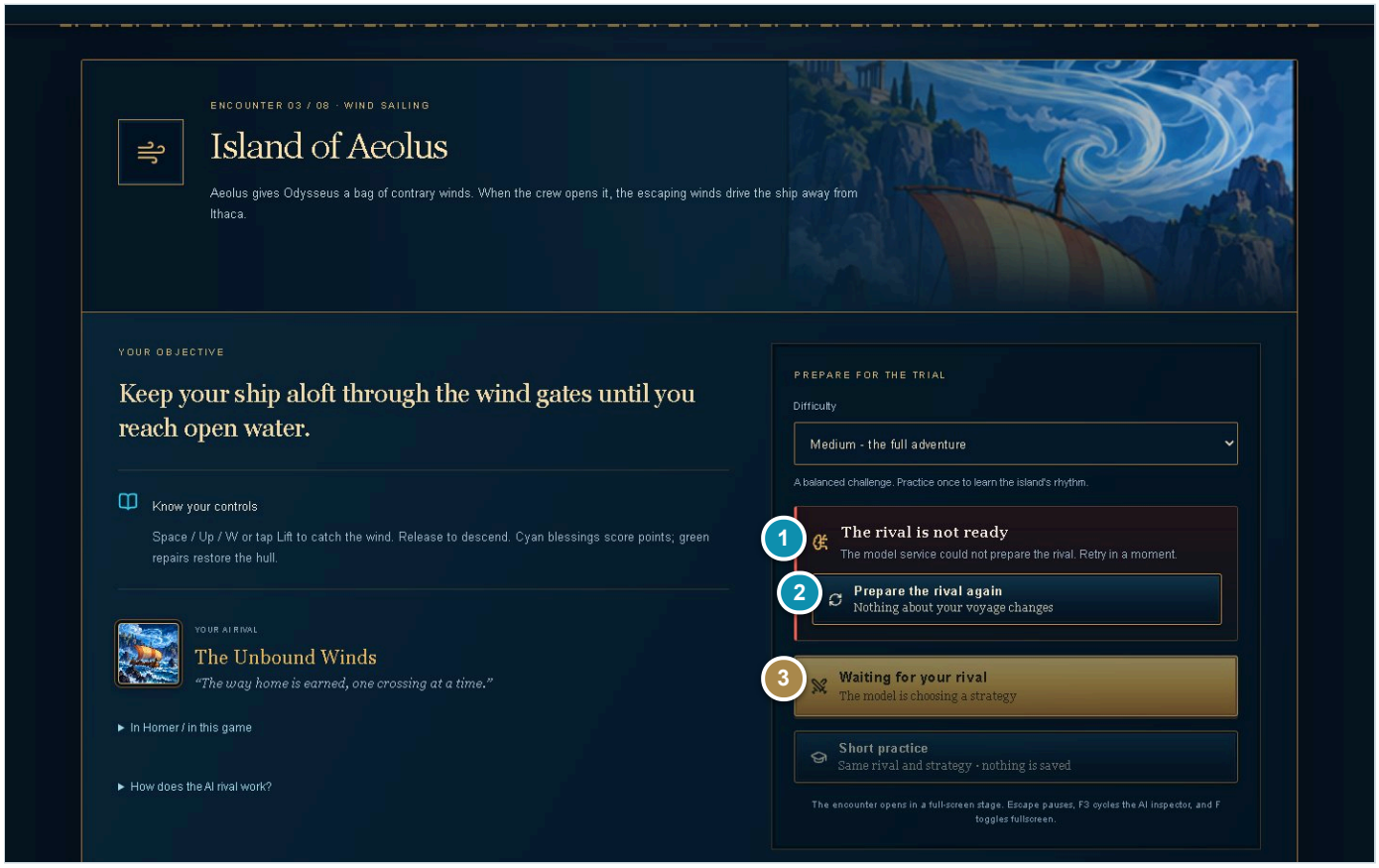


AI state: User control over the AI result: regenerate, start, or practise

1. Ask for another strategy: regenerate the AI result.
2. Start repeats difficulty and plan and says it counts toward the voyage. Enter is the shortcut.
3. Short practice: same rival and strategy, nothing saved.
4. "How does the AI rival work?" says what the model decides and what the game always owns (collisions, scoring, progress).
5. Explains Escape, F3 and F before the full-screen stage opens.

D09 Briefing: AI error (state 5 of 5)

F4 Recover from a failed AI plan (A1 error) · Rosa, Daniel

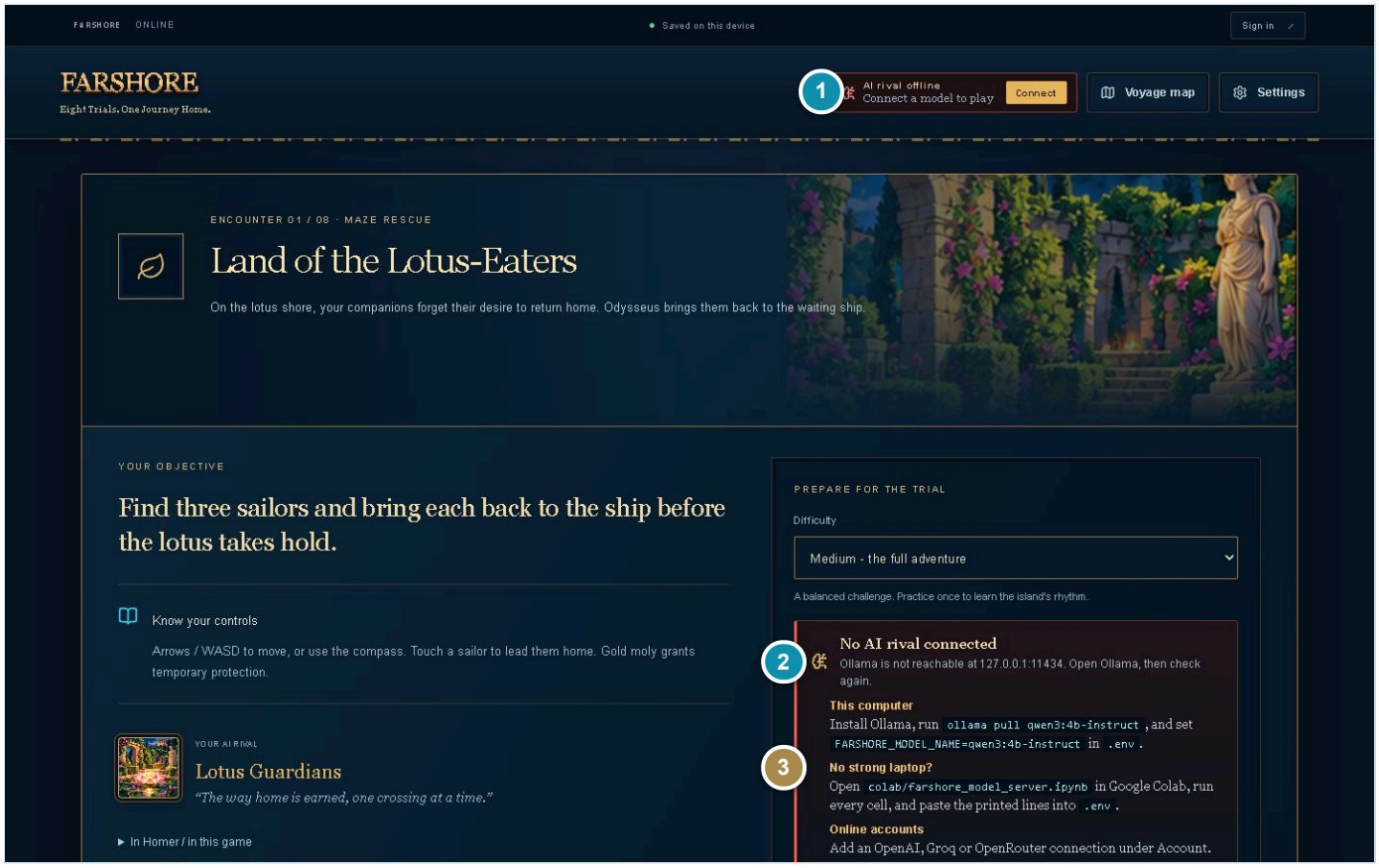


AI state: Error: the strategy request failed. Triggered with the mock AI service (mode error-once), which is how Task 4 of the test reaches it

1. Aplain message with no error codes, in the red error style.
2. One clear recovery action, "Prepare the rival again", plus the reassurance "Nothing about your voyage changes".
3. Start stays disabled. Pilot: its label "The model is choosing a strategy" is wrong while nothing is being prepared.

D10 Briefing: no AI rival connected (initial state without a model)

F3 Prepare the rival (A1 unavailable) · Daniel, Rosa



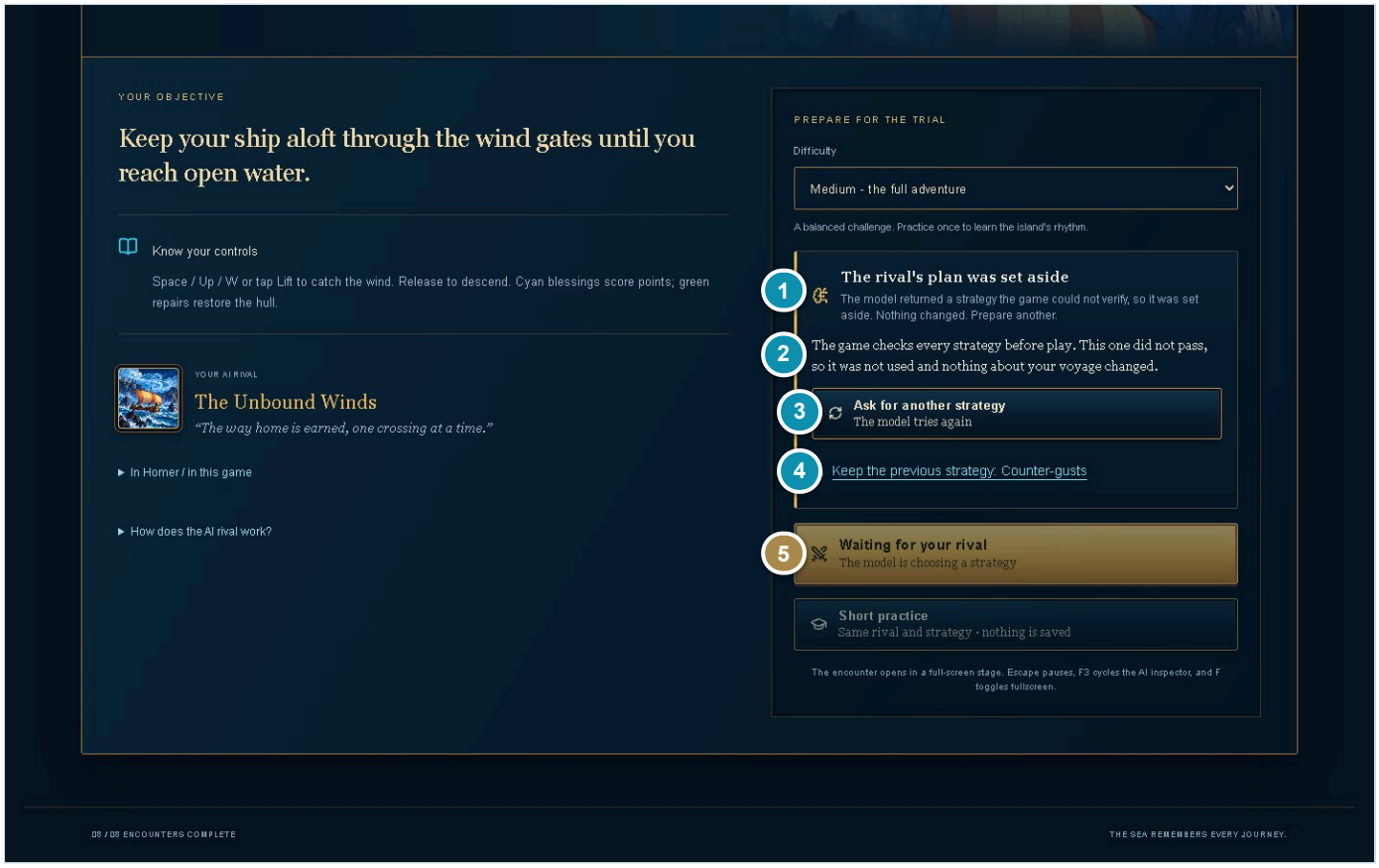
AI state: Model service unreachable: play is blocked until a model is connected

Failure simulated for this capture (the status check reported the model as unreachable (the mock AI service's offline mode produces the same screen)); the interface is real.

1. The header chip turns to "AI rival offline", with Connect.
2. Names the exact problem and the next step: open Ollama, then check again.
3. Three ways to connect: this computer, Google Colab, or an online account.
4. Check again sits below the fold. The game also re-checks every 15 seconds (T4).

D22 Briefing: low-confidence result (state 4 of 5)

F4 Recover from a failed AI plan (A1 uncertain) · Rosa, Maya

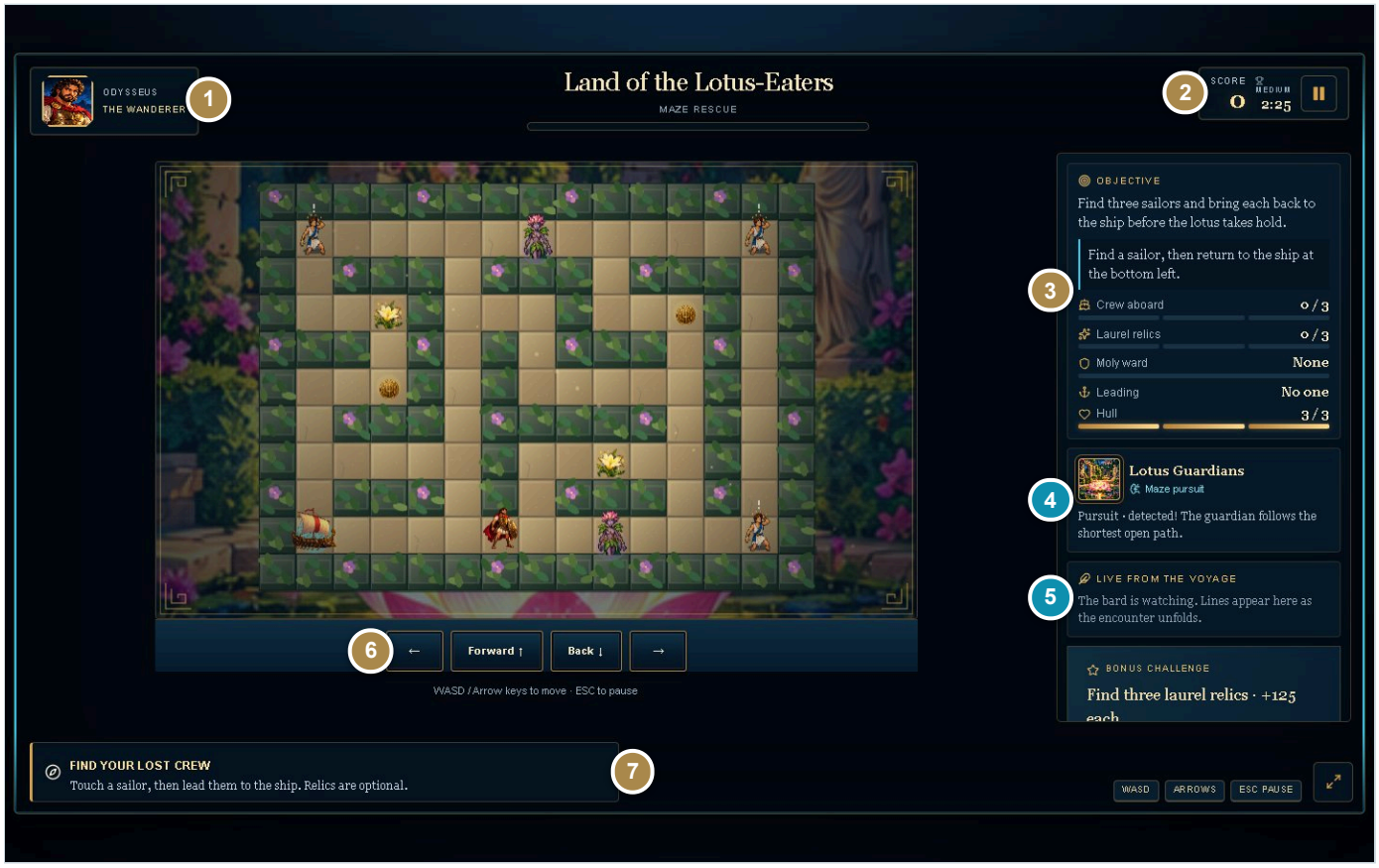


AI state: Low-confidence: the model returned a strategy that failed the game's checks, so it was set aside. Triggered with the mock AI service (mode uncertain-once) after the player asked for another strategy

1. Amber, not red: the AI produced an answer, but the game could not trust it.
2. Explains the check in plain words. Large language models give no reliable confidence number, so none is invented.
3. Ask for another strategy: the user regenerates the result. The rejected plan is never used.
4. Undo: Keep the previous strategy returns to the plan the player had before asking again.
5. Pilot issue 11: the disabled start button still says "The model is choosing a strategy" while nothing is being prepared.
6. New in Phase 3: before this, an unverifiable plan looked identical to a connection error.

D11 Lotus-Eaters: playing

F4 Play · Maya, Daniel

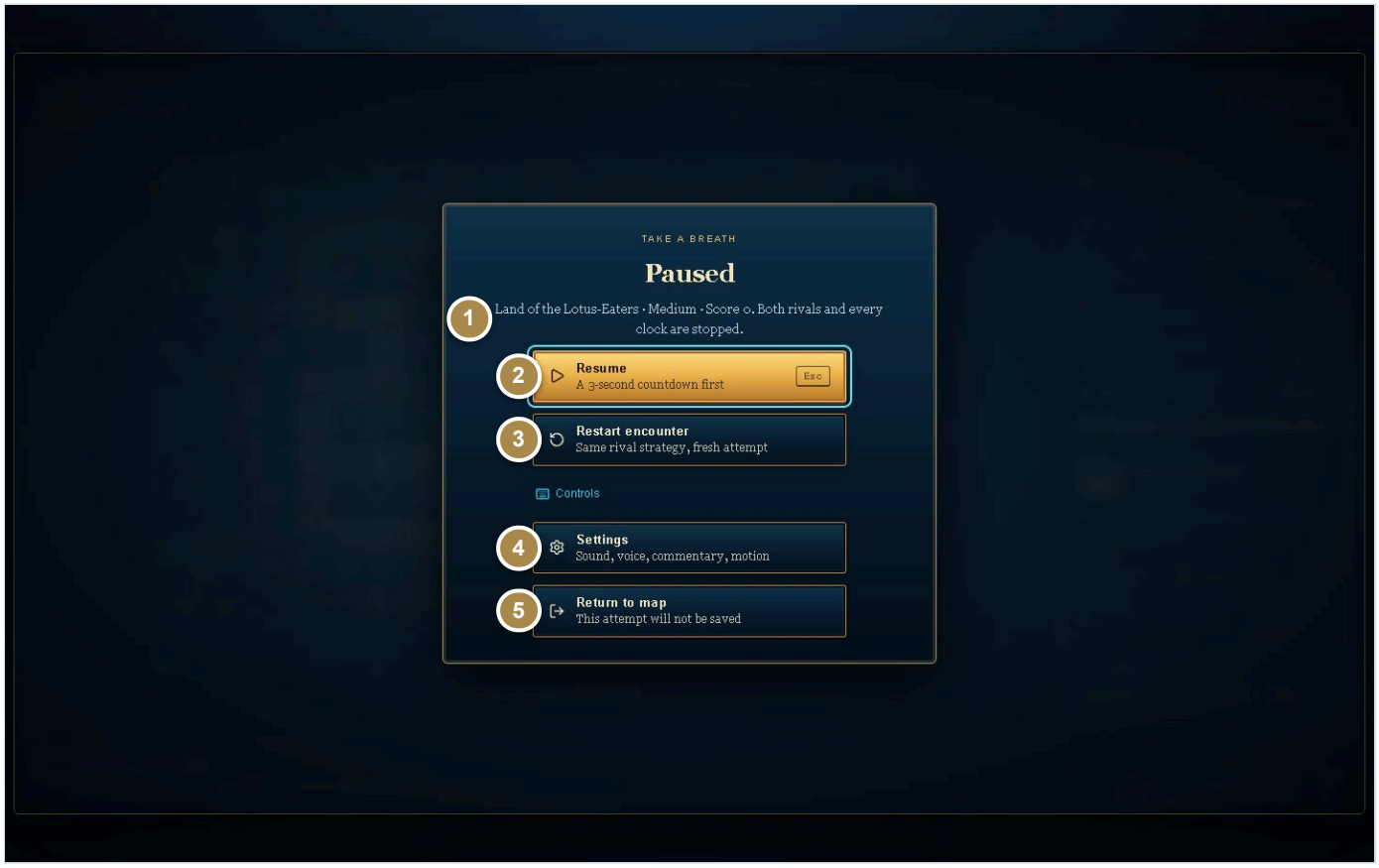


AI state: The chosen strategy runs locally every frame. Commentary lines are written by the model during play

1. Odysseus' portrait and lives, top left.
2. Score, difficulty, time and pause, top right.
3. Objective and progress counters stay in view.
4. Rival card: plan name and current behaviour. Pilot: the plan name and the live line can disagree.
5. Live from the voyage: the bard's and rival's lines appear here and are spoken.
6. On-screen direction buttons for mouse and touch players.
7. Tactical cue strip with the current instruction.

D12 Pause

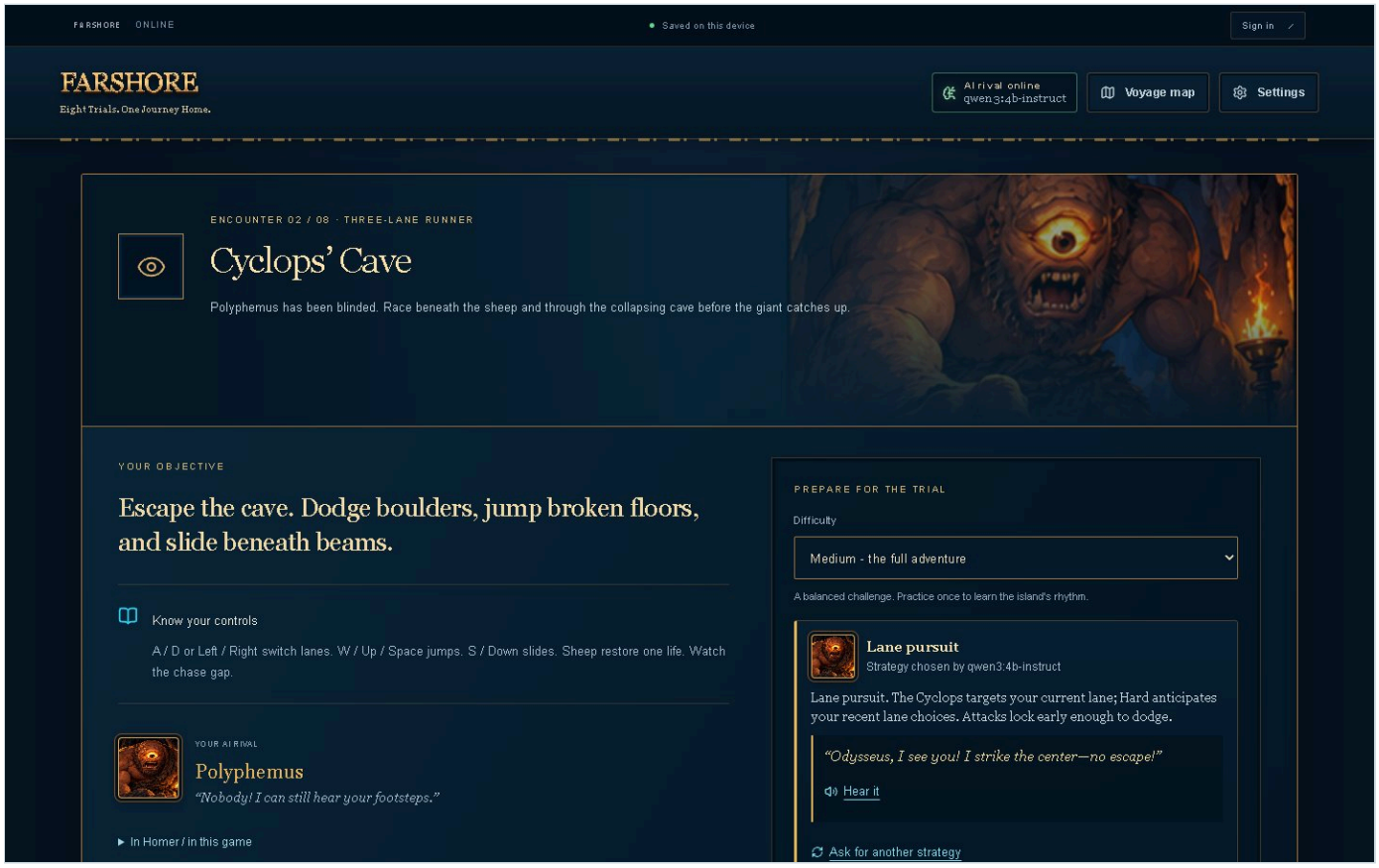
F4 Play · All players



1. States exactly what is paused: both rivals and every clock.
2. Resume has a 3-second countdown, so the player is not hit on return. Esc shortcut.
3. Restart keeps the same rival strategy.
4. Settings are reachable mid-play.
5. Return to map warns that the attempt will not be saved.

D13 Cyclops' Cave briefing

F3 Prepare the rival (A1 success) · Maya

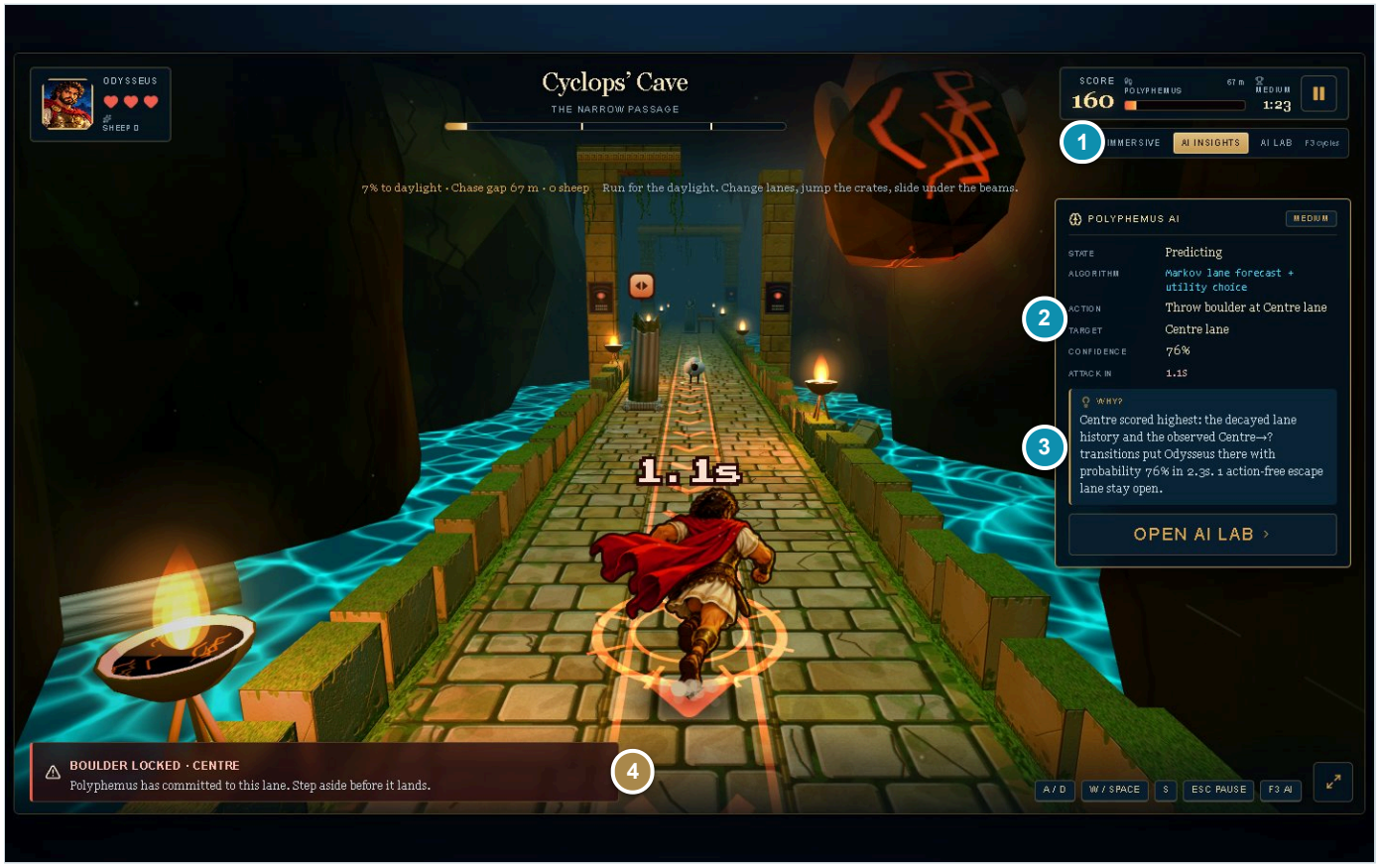


AI state: A1 success for Polyphemus

1. Same briefing layout for every trial, so learning carries over. The cave adds an Original cave tactics practice.

D14 Cyclops' Cave: AI Insights

F5 Understand the rival · Maya, Rosa

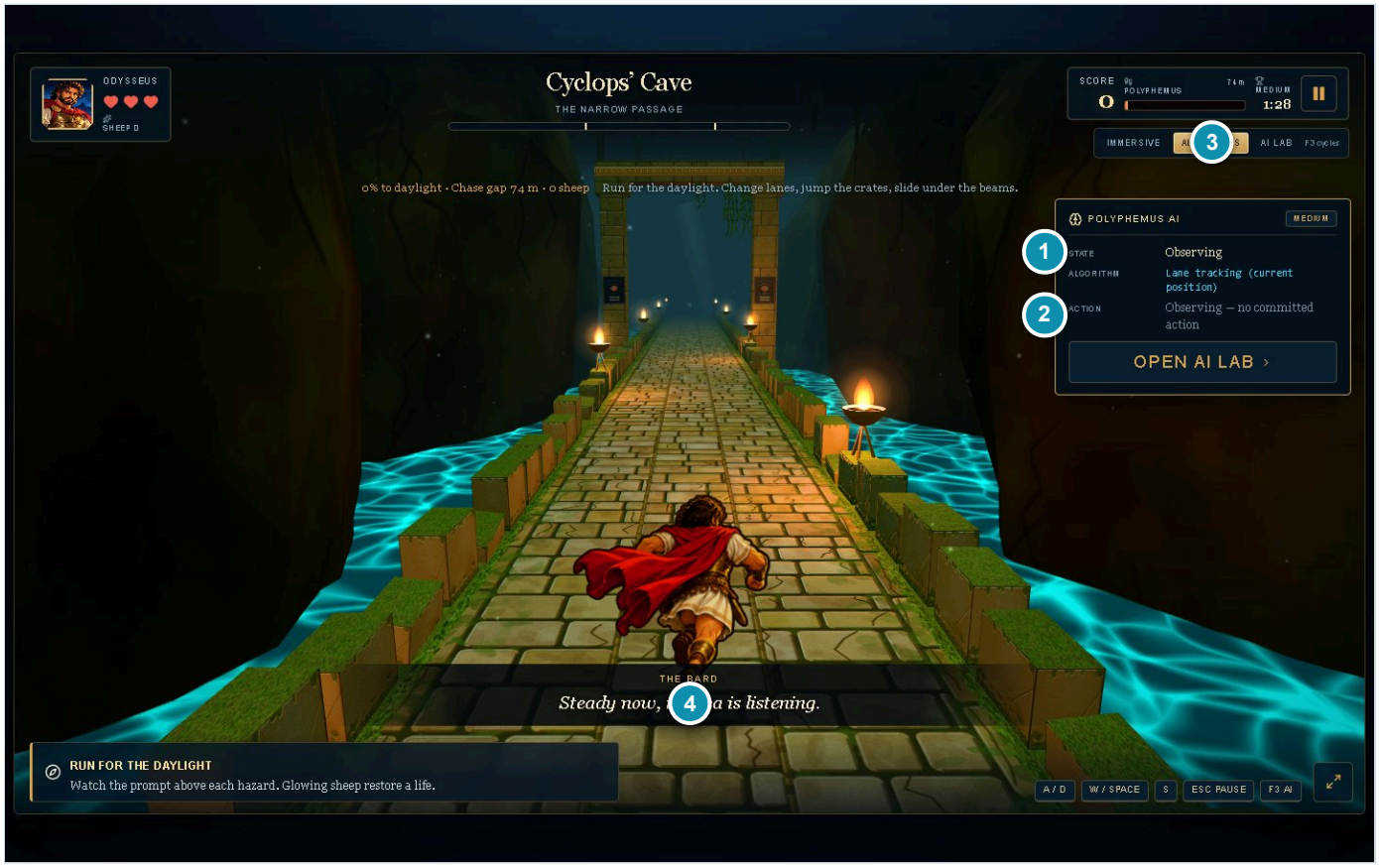


AI state: Explainable AI overlay: state, algorithm, chosen action and a grounded reason

1. Mode chips: Immersive, AI Insights, AI Lab. F3 cycles them.
2. State, algorithm, chosen action, target lane, forecast probability and time until the attack.
3. "Why?" gives the grounded reason, including the guarantee that an action-free escape lane stays open (T3).
4. The danger cue is red and in words: "Boulder locked, Centre. Step aside before it lands."
5. The 76% is a real Markov-model forecast of the player's lane, not an invented confidence. Cyan is used only for AI information.

D23 Cyclops' Cave: AI Insights before the first decision (state 1 of 5: empty)

F5 Understand the rival · Maya, Rosa

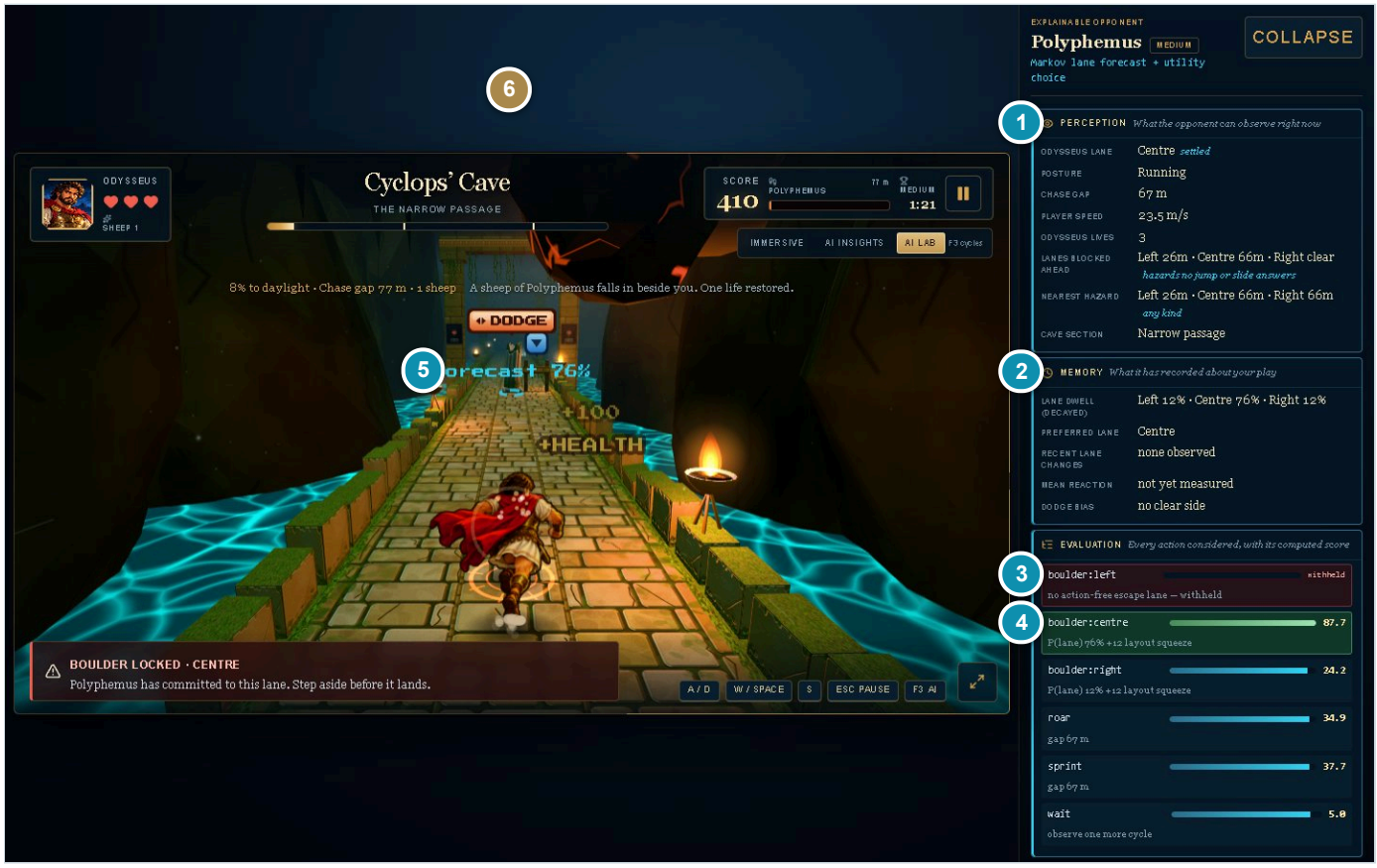


AI state: Empty state of the explainable AI panel: the rival is observing and has committed to nothing yet

1. State: Observing. The panel says what the rival is doing even before it acts.
2. "No committed action": an honest empty state, not a blank box or a made-up reason.
3. AI Insights is on by default in this encounter. F3 cycles Immersive, AI Insights and AI Lab.
4. Live bard commentary. In this capture the line comes from the mock AI service.

D15 Cyclops' Cave: AI Lab panel

F5 Understand the rival · Rosa

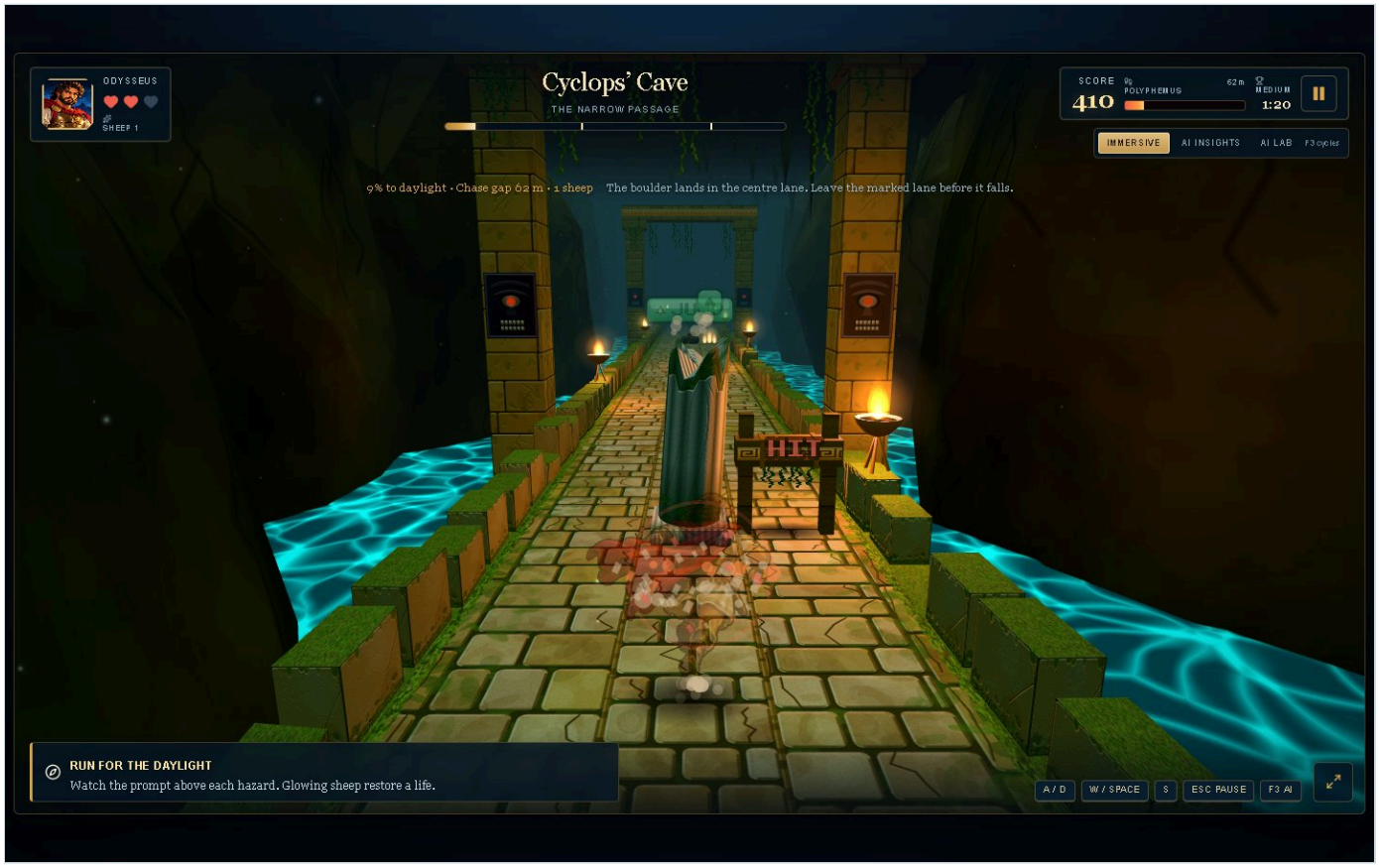


AI state: Full decision pipeline: perception, memory, evaluation, decision, execution, result

1. Perception: exactly what the opponent can observe right now.
2. Memory: what it has recorded about the player's habits (lane dwell, preferred lane).
3. A withheld option is shown dimmed, with its fairness reason (no action-free escape lane).
4. Every candidate action with its computed score. The chosen one is highlighted.
5. The forecast is also drawn in the scene over the targeted lane.
6. The play area shrinks to make room and returns to full width when the panel is collapsed.

D16 Cyclops' Cave: Immersive

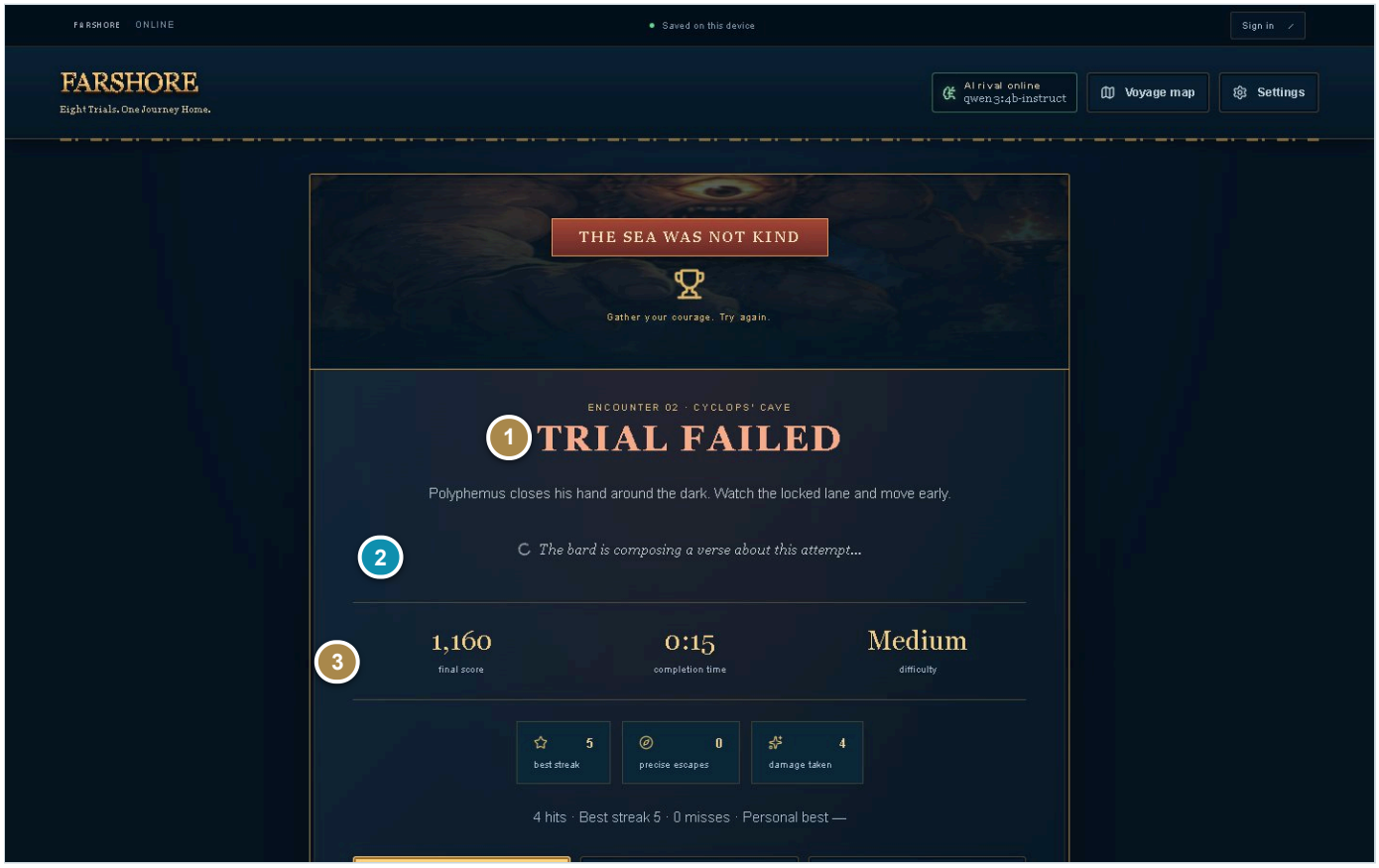
F4 Play · Maya



1 Immersive hides the inspector. Fair attack warnings (the locked-lane stripe and countdown) always stay, because they are gameplay information, not AI debugging.

D17 Result: trial failed

F4 Play · Daniel, Maya

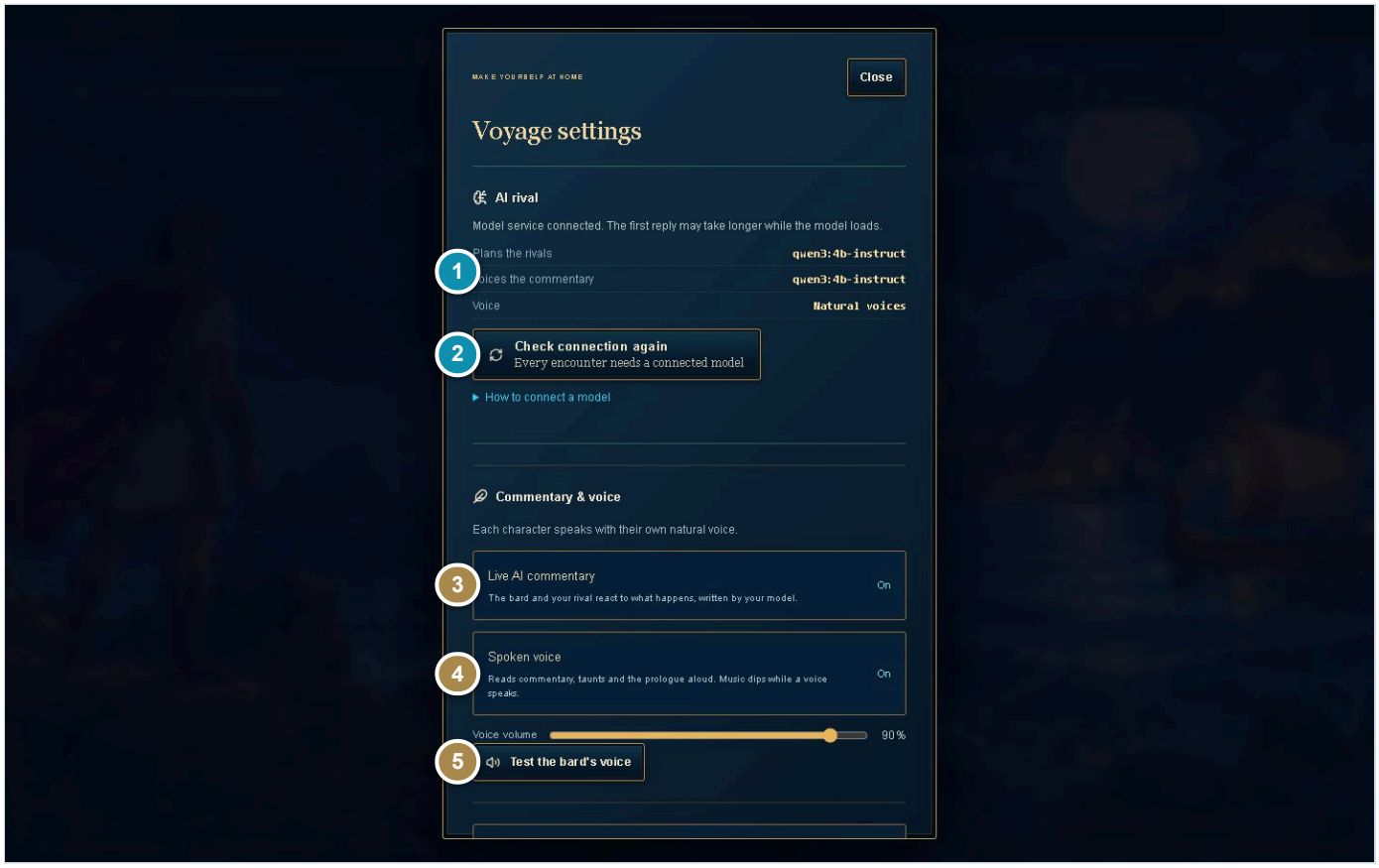


AI state: Model-written rival line reacting to the attempt

1. Clear outcome, with a specific tip: "Watch the locked lane and move early".
2. The rival reacts to this attempt in a line written by the model.
3. Score, time and difficulty.
4. Retry, Change difficulty, World map and an accessibility Assisted passage sit below.

D18 Settings

F6 Settings · Daniel, Rosa

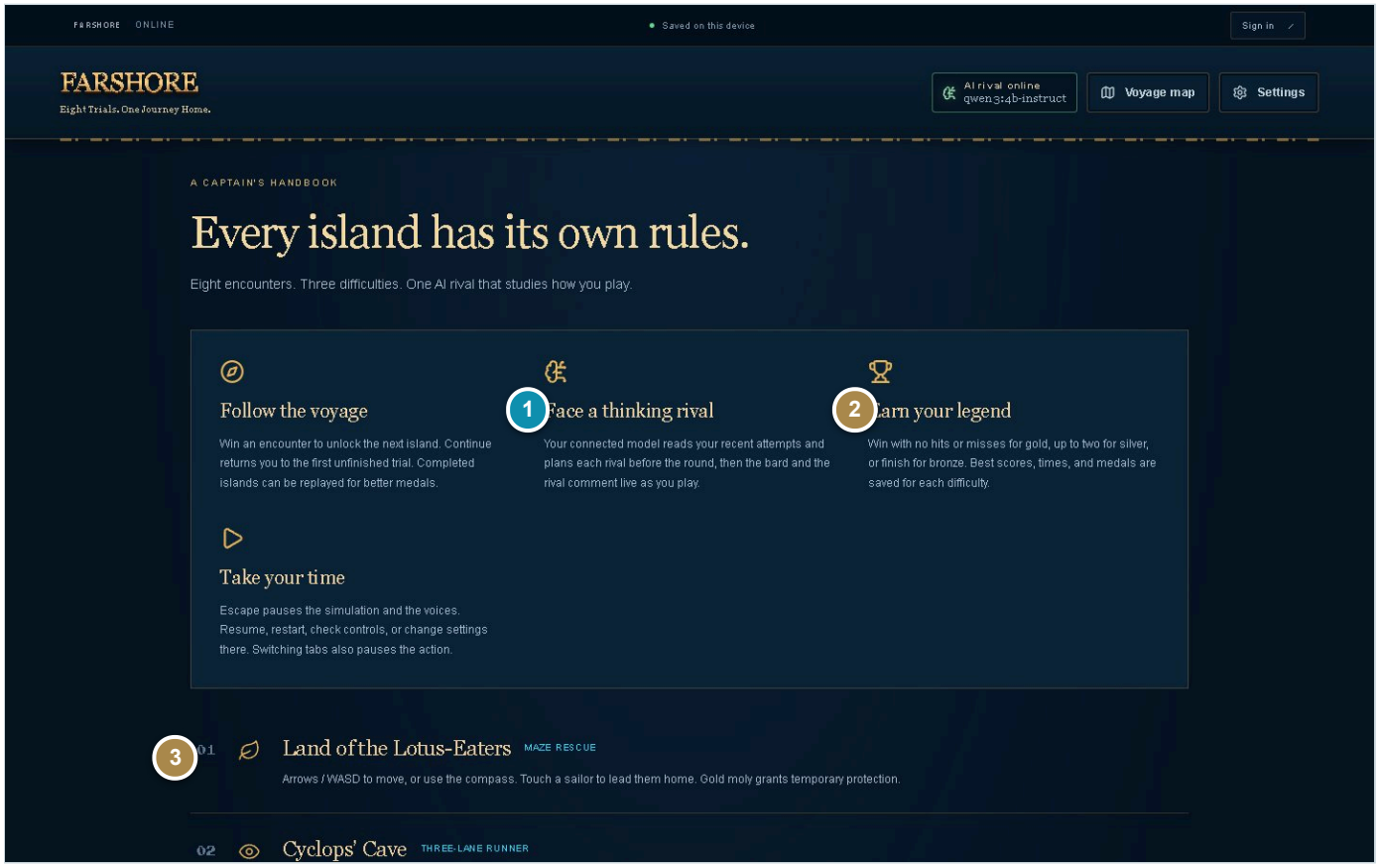


AI state: Which model plans the rivals and which voices the commentary

1. Names the models in use for planning, commentary and voice.
2. Check connection again, with the reminder that every encounter needs a model.
3. Live AI commentary on or off.
4. Spoken voice on or off, separately from commentary.
5. Voice volume and a voice test.

D19 How to play

F2 Learn · Daniel

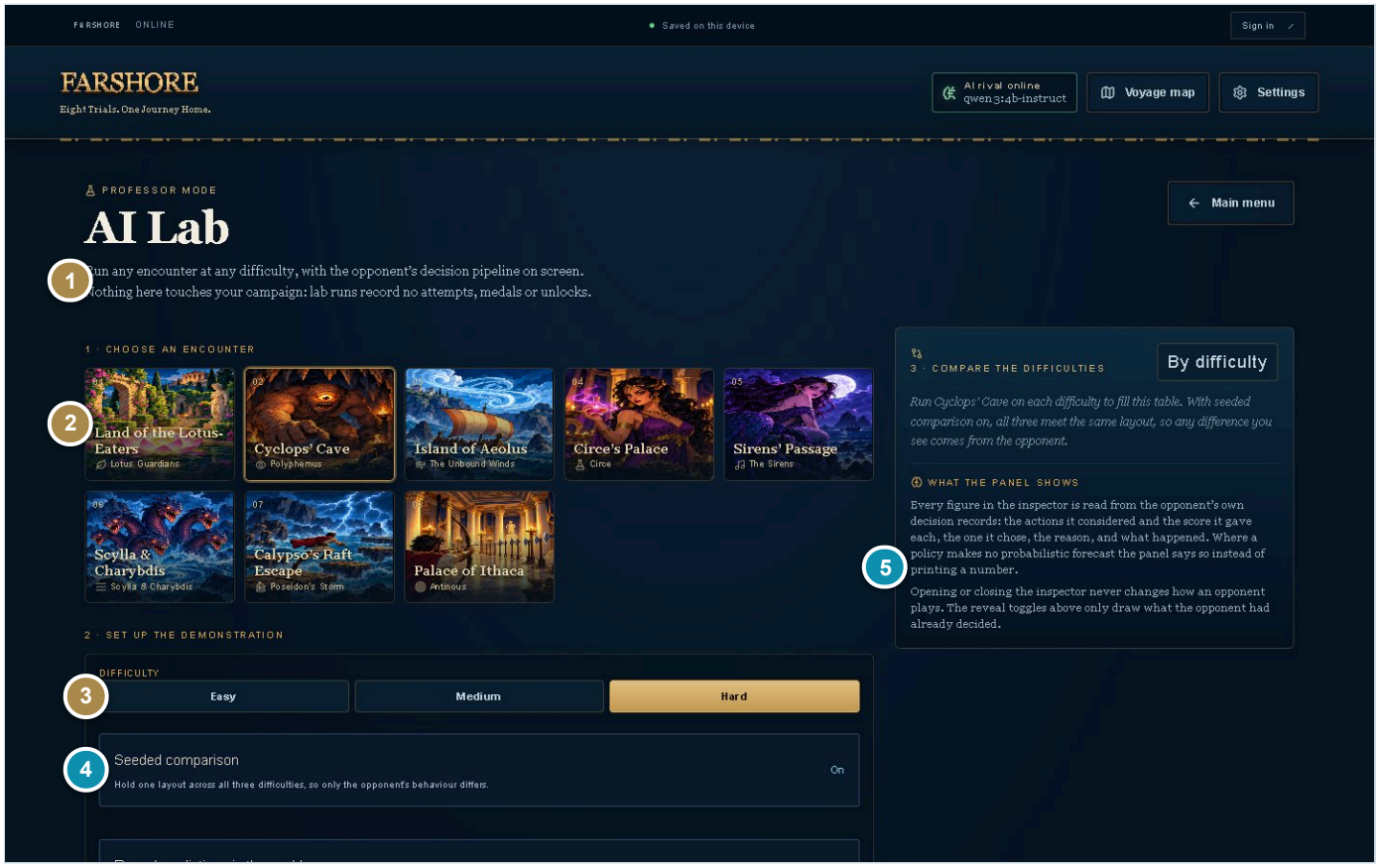


AI state: Explains the AI rival in one sentence

1. "Face a thinking rival": the model reads your attempts and plans the rival, then comments live.
2. The medal rules are stated exactly.
3. Controls for all eight trials on one page.

D20 AI Lab (professor mode)

F5 Understand the rival · Rosa

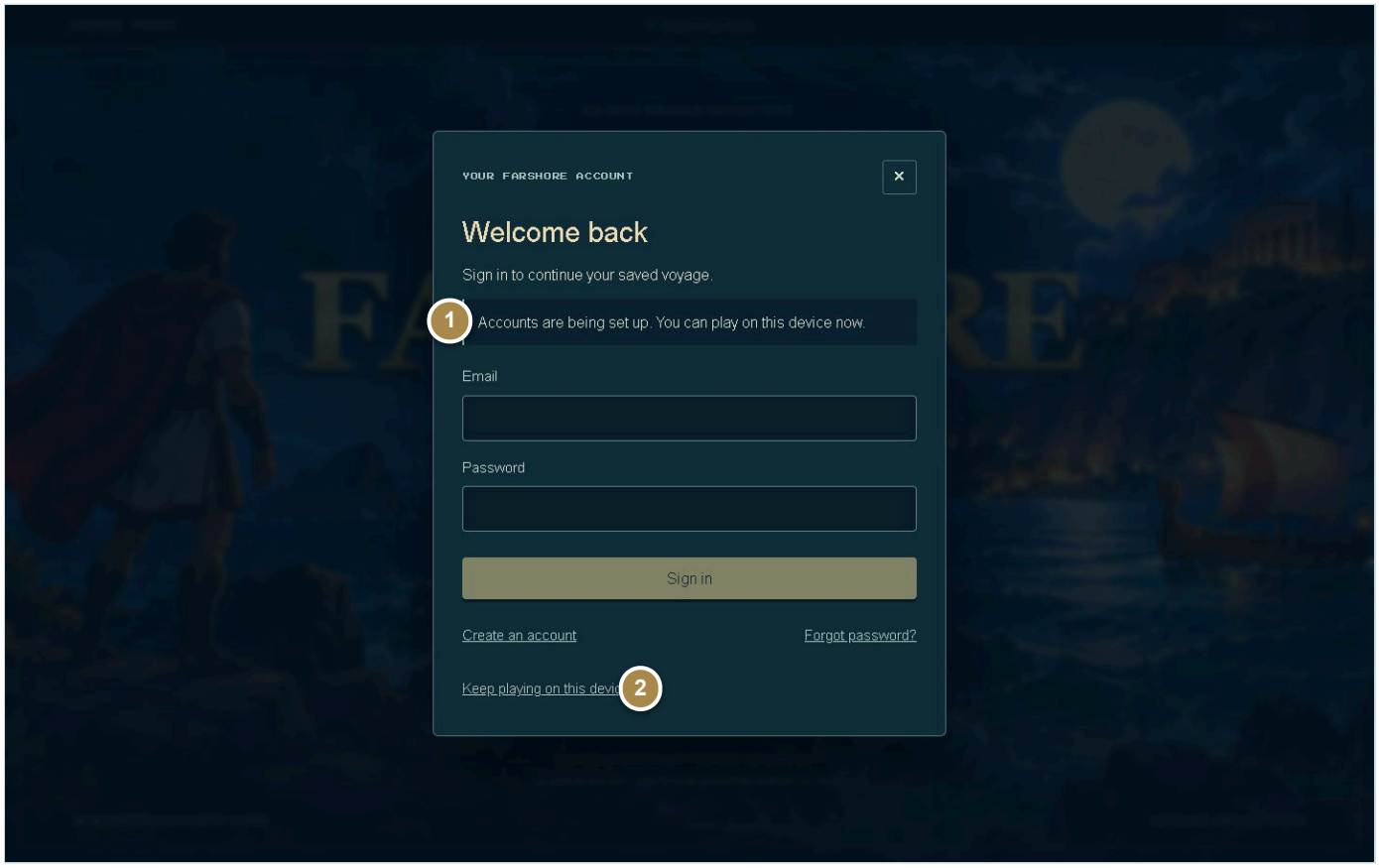


AI state: Compare the rival across difficulties with the same layout

1. States that lab runs never touch campaign progress.
2. Choose any of the eight encounters.
3. Choose a difficulty.
4. Seeded comparison keeps the layout fixed, so any difference comes from the opponent.
5. Says the inspector never changes how the opponent plays.

D21 Sign in

F7 Account · Maya

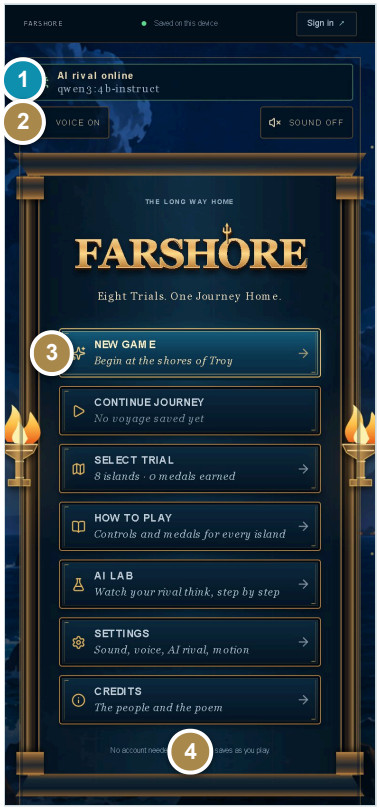


1. When accounts are not configured, says so and offers play on this device.
2. "Keep playing on this device" is always available.
3. Email/password and Google sign-in sync saves through MongoDB when the server is configured.

1.4 Phone screens (390 × 844)

M01 Title menu (phone)

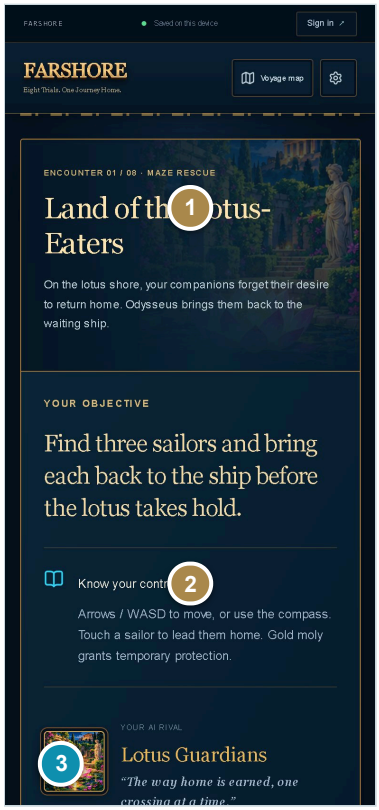
F1 Start · All players



1. The AI rival status gets its own full-width row at the top.
2. Voice and sound toggles follow directly below.
3. The menu stacks into one column, and every entry keeps its one-line explanation.
4. "No account needed. Your voyage saves as you play."

M02 Briefing (phone)

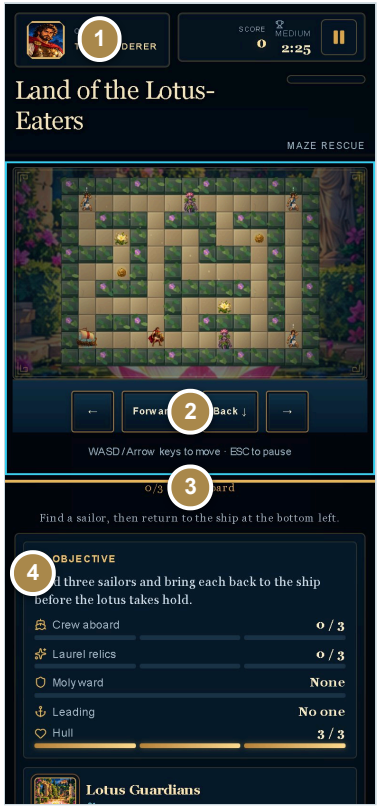
F3 Prepare the rival (A1 success) · Daniel



1. Story and objective come first, in large type.
2. Controls are explained in words, including touch.
3. Pilot: on a phone the rival's strategy panel and the start button are several screens further down.

M03 Lotus-Eaters (phone)

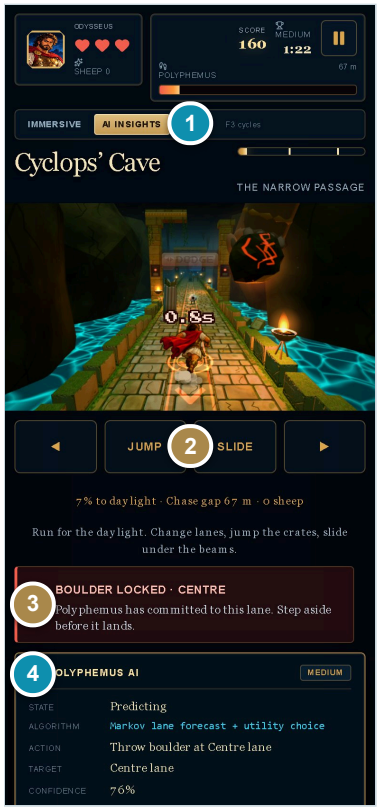
F4 Play · Maya, Daniel



1. Portrait, then score, time and pause: the same HUD order as desktop.
2. Large on-screen direction buttons for touch.
3. Progress (crew aboard) and the current instruction sit directly under the maze.
4. Objective and counters follow below.

M04 Cyclops' Cave with AI Insights (phone)

F5 Understand the rival · Maya, Rosa



AI state: Insights panel below the play area

1. Mode chips: Immersive, AI Insights, AI Lab.
2. Touch controls: lane left and right, Jump, Slide.
3. The locked-lane warning is in words as well as colour.
4. Pilot: the Insights panel sits below the play area on a phone, so it cannot be read while dodging.

2. Test script

The instructor's Usability Test Script Template, filled in for Farshore (`Farshore-Usability-Test-Script.docx`). The same script is used for every participant, and Phase 5 repeats it for the before-and-after comparison.

Session information

Field	Fill in
Team / app name	Farshore (solo project, Bibek Pathak)
Participant ID	P1, P2, ... (per session)
Date and time	per session
Moderator / note-taker	Bibek Pathak (moderator and note-taker)
Build tested	Farshore 3.0.0, Nov 5 2026 build, run locally. Tasks 1-3 use Qwen3-4B-Instruct through Ollama; Task 4 uses the mock AI service

Setup before each session (about 3 minutes)

1. Open the Ollama app and run `npm.cmd run dev`. Open http://localhost:3000, wait for **AI rival online**, and open one trial briefing yourself so the model is warm.
2. Open a **new InPrivate (Edge) or Incognito (Chrome) window** at http://localhost:3000, so each participant starts a fresh game. Press F11.
3. Open `session-logger.html` on a second screen or device, or use a printed copy of the script. Enter the participant ID.
4. Have the four task cards and the SUS form ready.

1. Welcome and consent (read aloud)

"Thank you for helping us. We are students in an HCI course testing an app we built that uses AI. We are testing the app, not you. If something is confusing, that is a problem with our design, and that is exactly what we need to find.

I will ask you to do a few tasks. Please think out loud as you go: say what you are looking for, what you expect to happen, and anything that surprises you. I will not be able to help you during the tasks, because we need to see what happens without us.

This will take about 20 minutes. Your name will not appear in our report. You can stop at any time, and you can skip any task, with no explanation needed. Please do not enter any personal or sensitive information into the app.

Do you have any questions before we start?"

Record: consent given (yes/no). Sessions are not recorded unless the participant agrees.

2. Warm-up questions (2 minutes)

1. How often do you play video games, and what kind?
2. When you play against a computer opponent, how do you work out what it is planning?
3. Have you used AI tools (such as ChatGPT) or played games with AI-driven characters before?

3. Tasks

Read each task aloud and hand over the printed card. Start timing when they begin. Stop when they say they are done or give up. Record: completed (Yes / Partly / No / Gave up), time on task, errors or wrong turns, and quotes. After Tasks 1-3, ask: "How easy or difficult was that, from 1 (very difficult) to 5 (very easy)? Why?"

Task 1, from primary flow F1 (start or resume and learn a trial)

- **Task wording:** "You've heard about a game where you travel home with Odysseus, and you've just opened it for the first time. Start a new journey and get ready for the first challenge."
- **Success criterion:** the Land of the Lotus-Eaters briefing is on screen (it shows "YOUR OBJECTIVE").

Task 2, from primary flow F2 (prepare, review and accept a model strategy)

- **Task wording:** "Before you face your first opponent, you want to know what you're up against. Find out what your opponent plans to do in this challenge, and who or what decided that plan."
- **Success criterion:** they paraphrase the strategy in the rival panel **and** say the AI model chose it ("Strategy chosen by qwen3:4b-instruct").

Task 3, from primary flow F3 (play against an active mythic rival)

- **Moderator setup (untimed):** Settings > Unlock all islands, close Settings, open the Voyage map. Say: "I've unlocked another island to save time."
- **Task wording:** "A friend told you the Cyclops' Cave is the toughest island, and that you can see how the Cyclops thinks. Play it, and while you play, tell me why the Cyclops made one of his moves, using what the game shows you."
- **Success criterion:** they cite a reason or action from the AI Insights (or AI Lab) panel for a specific throw.
- **Also record:** the game result, and whether they noticed the AI Insights panel without prompting.

Task 4, the AI failure task, from primary flow F4 (recover from a failed or unsuitable AI plan)

- **Moderator setup:** in a second tab of the same browser window, open http://localhost:3000/api/ai-mock?mode=error-once, then close that tab. Do not warn the participant.
- **Task wording:** "You'd like to try a different island. Open the Island of Aeolus and get ready to play it."
- **What the AI does:** an error. The mock AI service fails the first strategy request, so the briefing shows "The rival is not ready" with "Prepare the rival again". The retry succeeds.
- **Record:** did they notice something was wrong (yes/no)? What did they do next? Did they recover without help (yes/no)? Success means they reach a ready rival (the start button reads "...counts toward your voyage") without help.
- After the session, open http://localhost:3000/api/ai-mock?mode=off.

4. Moderator rules

Do	Do not
Stay quiet and let silence sit	Explain where to click or what a button does
Ask "what are you thinking?" when they go quiet	Say "it's over there" or point
Ask "what did you expect to happen?"	Defend the design or explain your reasoning
Ask "what would you do next if I weren't here?"	React with "yes!", "no" or "exactly"
Write down the exact words they use	Ask leading questions ("wasn't that easy?")
Let them struggle for at least 60 seconds	Finish their sentences or guess their intent

If they are completely stuck and distressed, you may end the task. Record it as not completed and note where they stopped.

5. System Usability Scale (after all tasks)

The participant rates each statement from 1 (strongly disagree) to 5 (strongly agree). Do not explain or reword the statements. If asked, say "whatever it means to you."

1. I think that I would like to use this app frequently.
2. I found the app unnecessarily complex.
3. I thought the app was easy to use.
4. I think that I would need the support of a technical person to be able to use this app.
5. I found the various functions in this app were well integrated.
6. I thought there was too much inconsistency in this app.
7. I would imagine that most people would learn to use this app very quickly.
8. I found the app very awkward to use.
9. I felt very confident using the app.
10. I needed to learn a lot of things before I could get going with this app.

Scoring: for odd-numbered items subtract 1 from the score; for even-numbered items subtract the score from 5. Add the ten adjusted values and multiply by 2.5. The result is 0-100, and about 68 is average.

6. AI trust and perception questions

1. On a scale of 1 (not at all) to 5 (completely), how much did you trust the results the app gave you? Why?
2. How did you know whether a result was good or not?
3. When the app had trouble, what did you think had happened?
4. Was there anything you wanted to change about a result but could not?
5. Would you use this instead of how you do it today? Why or why not?

7. Debrief

"That's everything. Two last questions: what was the most frustrating moment, and what worked best? Is there anything you expected the app to do that it didn't?"

Thank them. Do not explain what they "should" have done, and do not justify the design.

After the session

1. Within 10 minutes, write your top 3 observations while they are fresh.
2. In the logger, click **Export JSON** and save it as `phase3/data/P#.json`. This fills the results spreadsheet: completion, time on task, ease, SUS score and trust rating.
3. Do not change the script between participants. Note any change you were forced to make, and why.
4. Keep the completed script. Phase 5 reuses the same tasks and measures.

Analysis plan

- Per task: success rate (Yes), completion including Partly, median completed time, mean errors, and mean ease (1-5).
- SUS per participant, plus mean and standard deviation, compared with the 68 average. AI trust: mean, median and share rating 4 or 5, with the reasons.
- Task 4: share who noticed the failure and share who recovered without help.
- Issues are rated 0 (not a problem) to 4 (usability catastrophe) by frequency, impact and persistence. The most severe are fixed before the midterm and shown before and after. The rest are listed for Phase 4.

Materials used with this script: task cards, the SUS and AI trust questionnaire, the observer sheet and the session logger (`phase3/kit/`).

3. Test report

3.1 Method

Moderated, in-person think-aloud sessions with five or six adults who had not seen Farshore, run by the author with the script in Section 2 (the instructor's template, filled in). Each session took about 20 minutes and had four tasks (T1-T4), on a Windows laptop running the prototype locally. Tasks 1-3 used Qwen3-4B-Instruct through Ollama. Task 4 used the mock AI service to trigger an AI error without warning. Measures, as on the template: completion (Yes / Partly / No / Gave up), time on task, errors or wrong turns, ease from 1 to 5 after Tasks 1-3, and for Task 4 whether the participant noticed the failure and recovered without help. After all tasks: the System Usability Scale, and the template's five AI trust and perception questions (trust in the results, 1-5, with the reason).

Analysis: success rate per task (Yes only, and including Partly), median time on task, mean ease, SUS scored by the standard method and compared with the 68 average, mean AI trust, and an issue list rated on the 0-4 severity scale (0 = not a problem, 4 = usability catastrophe). All calculations are live formulas in the raw data workbook. These results are the baseline that Phase 5 repeats.

3.2 Participants

Pending real session data. Participant count and background (age range, gaming frequency, AI chatbot use, familiarity with the Odyssey). This section is generated from `Farshore-Phase-3-Test-Data.xlsx` after the usability sessions. No results are estimated.

3.3 Task results

Pending real session data. Completion, time, errors, assists and SEQ for each task. This section is generated from `Farshore-Phase-3-Test-Data.xlsx` after the usability sessions. No results are estimated.

3.4 SUS and AI trust

Pending real session data. SUS per participant and overall, and AI trust item means. This section is generated from `Farshore-Phase-3-Test-Data.xlsx` after the usability sessions. No results are estimated.

3.5 Findings

Pending real session data. Issues with severity, frequency and participant quotes. This section is generated from `Farshore-Phase-3-Test-Data.xlsx` after the usability sessions. No results are estimated.

3.6 Pilot walkthrough

Before the sessions, the task path was checked in an automated browser to confirm that every task can be completed and the script matches the interface. These are not participant results.

On 2026-11-04 the moderator's assistant ran the task path in an automated headless Edge browser (software rendering, so slower than a real laptop), with local Ollama `qwen3:4b-instruct`. This checks that each task can be completed and that the script matches the real interface. **It is not a usability result and is not counted in the data.**

Task feasibility

Task	Can it be completed?	Script consequences
T1	Yes. New game, then "The journey ahead", then "Land of the Lotus-Eaters" (or "Open voyage map", then card 1, then Enter trial) reaches the briefing.	Both paths count as success.
T2	Yes. The rival panel shows the strategy name, "Strategy chosen by qwen3:4b-instruct", a description and a taunt. Preparation took about 40 s in the headless run.	The 3-minute limit is enough. Warm the model before each session so the first plan is not a cold start.
T3	Yes, after unlocking (the moderator uses Settings > Unlock all islands). AI Insights is on by default, and only in the cave. Its first committed decision takes a few seconds of play.	Keep T3 on Cyclops' Cave. The participant needs to survive a few seconds before the panel shows a reason.
T4	Yes. With the mock AI service in mode `error-once`, the Aeolus briefing shows "The rival is not ready", and "Prepare the rival again" then succeeds. Checked in the browser on 2026-11-04.	The failure is triggered with the mock service, as the brief requires, instead of quitting Ollama.

The protocol follows the instructor's template and was cut from six tasks to four (about 20 minutes per session). The dropped tasks were playing a full Lotus round on Easy and finding the voice toggle in Settings. Both completed in the pilot. Observations 4 and 5 below come from them.

Things to watch for during sessions (pilot observations, to be confirmed or not by participants)

1. **The start button is below the fold.** At 1440 × 900 the briefing's "Face [rival]" button sits below the visible area, so players must scroll past the objective and the rival panel to start (T1, T2).
2. **The Lotus strategy has no gameplay effect.** In the Lotus-Eaters maze, the guardians' behavior depends only on difficulty. The strategy the model chooses (Garden patrol / Maze pursuit / Junction interception) is shown, but the maze code never reads it. This matters for T2 and AT1 to AT3 (AI honesty). Recommendation: either make the plan change guardian behavior or say so in the panel. **Decision (2026-11-04): tested as-is.** Watch whether participants notice in T2 and T3. It will be fixed after the sessions and reported as a change.
3. **The plan name repeats and the panel disagrees with the HUD.** The rival panel title "Maze pursuit" is repeated as the first words of its description. In play, the HUD says "Maze pursuit" under the rival but "Patrol · guardians follow the garden paths" as the live line. Players may read these as conflicting (T2).
4. **The Lotus side panel is clipped.** At 1440 × 900 the "Bonus challenge" card in the encounter side panel is cut off at the bottom.
5. **Medium is the default difficulty.** New players may not notice the selector (not a task in the 20-minute protocol).

6. **On a phone, AI Insights sit below the play area.** In Cyclops' Cave at 390 × 844 the Insights panel is under the game and touch controls, so it cannot be read while dodging. Sessions run on a laptop, so this is a design note rather than a test task.
7. **On a phone, the briefing is long.** The rival's strategy and the start button are several screens below the objective.
8. **The voyage chart labels overlap.** Some island labels overlap the "wine-dark sea" caption at 1440 × 900.
9. **Model availability can flicker under load.** Once during the automated run, the status check briefly reported "AI rival offline" while the laptop was busy, then recovered by itself. If it happens in a session, note it. It may affect T2 waiting times.
10. **"Cancel preparation" starts again straight away.** Cancelling returns the panel to its initial state, and the briefing then starts a new preparation by itself, so cancelling only restarts the wait. Watch whether anyone tries it in T2.
11. **The start button says "The model is choosing a strategy" when nothing is being prepared.** In the error and low-confidence states the disabled start button keeps that label, which can mislead (T4).

4. List of changes

Part A lists what changed between the Phase 2 design (the October 8 arcade package, 15 screens and wireframes W01-W28) and the prototype that was tested. Part B lists changes made because of the usability test. Part B is completed only after the sessions, from the Issues sheet in the test data workbook.

Phase 2 personas are referenced as Maya (a gamer who cares about fairness), Daniel (a reader with limited technical comfort) and Rosa (skeptical of AI). They remain the Phase 1 sketches, not new research.

Part A. Phase 2 design to tested prototype

ID	Change	Why	Replaces in Phase 2	Evidence
A1	Five trials became eight: Lotus-Eaters maze rescue, Cyclops' Cave runner, Aeolus wind sailing, Circe's potion duel, Sirens' melody memory, Scylla and Charybdis paddle defense, Calypso's raft hopping, Ithaca archery.	Follows the Odyssey route more closely and gives each episode its own recognizable game. Lotus-Eaters, a maze with simple movement, now comes first.	S03 map with five shores; S05-S09 rounds	`src/lib/arcade/arcade.ts`, commits efcce87-1b8d804
A2	The AI rival is no longer optional. Every encounter needs a connected model, and there is no model-free mode.	The AI opponent is the core experience, not an add-on. A clear "AI rival offline" state with connection steps replaces the optional toggle.	S04 briefing "mode" choice and A1 optional planning	`src/ArcadeApp.tsx` rival panel; decision recorded 2026-10-26
A3	The model changed from Llama 3.2 3B to Qwen3-4B-Instruct-2507 (Ollama `qwen3:4b-instruct`). Online accounts can use OpenAI, Groq or OpenRouter instead.	Better instruction and JSON-schema following (MMLU-Pro 69.6 vs 22.2) and no slow thinking phase, which matters for live commentary. Hosted providers help players without a strong laptop.	Phase 2 spike model	README "Connect the AI rival"
A4	Live AI commentary was added: a Homeric bard and the rival comment on play, and the bard writes a short verse after each attempt. Lines are spoken with Kokoro-82M neural voices, with the browser voice as the fallback.	Makes the AI's presence felt during play rather than only before it. Players can switch commentary and voice off separately in Settings.	Not in Phase 2 (model was used only before a round)	commit fbf5cf2; `src/lib/arcade/commentary`

ID	Change	Why	Replaces in Phase 2	Evidence
A5	Explainable AI inspector: AI Insights (default) and AI Lab views over Cyclops' Cave, cycled with F3 or on-screen chips. They show the rival's state, scored candidates, the chosen action, a grounded reason and a fairness guarantee.	Directly answers the Phase 2 trust question: players (Maya, Rosa) can see why the rival acted and that it cannot make an unfair attack.	Phase 2 showed only the plan name before a round	commits d6db373, f9067cd; `src/components/ai/ai-inspector.tsx`
A6	AI Lab (Professor) mode from the main menu runs any encounter at any difficulty, side by side, without touching the campaign save.	Lets an instructor or a curious player compare rival behavior. Supports the class demonstration.	No Phase 2 equivalent	commit d2d0301
A7	Gameplay moved from a canvas inside the page to a full-viewport 16:9 stage, with F for fullscreen and Escape to pause, and HUD overlays.	The highest-priority item in the redesign brief: the small embedded canvas competed with page chrome for space.	S05-S09 layout, S13 pause	`src/components/play/game-stage.tsx`, `docs/UI_DESIGN.md`
A8	Cyclops' Cave became a 3D three-lane runner (three.js), and Ithaca became a 3D great hall for the archery contest. A 2D fallback is kept for devices without WebGL.	Stronger sense of place and clearer depth cues for dodging and aiming.	S05 grid stealth, S09 archery	commits 4090a08, 14ae287
A9	Online accounts: email/password and Google sign-in, MongoDB saves across devices, and private per-account AI provider settings. Guests still save on the device.	Lets a signed-in player continue on another device and use a hosted model without a strong laptop. Provider keys stay server-side.	Phase 2 browser-local save only	commits efcce87, 4de1cfe; `SECURITY.md`
A10	Sound rebuilt: per-scene ambience and reverb, and physical-model sound effects that follow game events. Music dips while a voice speaks.	Audio cues reinforce on-screen warnings without replacing them. Every cue is also visible.	Phase 2 optional simple tones	commit 3981e36; `src/lib/audio/`
A11	Briefing redesign: objective, controls, rival portrait and quote, "In Homer / in this game", "How does the AI rival work?", difficulty with a plain-language note, the rival panel and a short practice that saves nothing.	Keeps Daniel's need for a clear objective and practice, and states plainly what the AI decides and what the game decides.	W04-W09 briefing wireframes	`src/ArcadeApp.tsx`

ID	Change	Why	Replaces in Phase 2	Evidence
A12	The voyage map gained a trial gallery and an illustrated voyage chart, with a per-trial AI technique note.	Faster trial choice, and the AI technique is shown before the player commits.	S03 map	`src/components/menu/`
A13	Settings now group AI rival status (models in use, check connection, how to connect), commentary and voice, sound, motion and visual effects, earlier modes, and a showcase "Unlock all islands".	One place to understand and control the AI and accessibility options.	S12 settings overlay	`src/ArcadeApp.tsx` settings dialog
A14	A mock AI service sits behind the single module that makes every model call. It can return success, a slow response, a low-confidence result, an error or "offline", switched from a second browser tab during a session.	The brief asks for every AI state to be shown and tested on demand. Task 4 uses it to lead participants into an AI error without warning.	Phase 2 could show failure states only by stopping the model	`server/lib/mock-ai.ts`, `app/api/ai-mock/route.ts`, `tests/mock-ai.test.mjs`
A15	A separate low-confidence state, "The rival's plan was set aside". It is amber, not red, and its main action is to ask for another strategy.	Before this, a strategy that failed the game's checks looked the same as a connection error. Rosa needs to see that the game refused to use a doubtful result.	Phase 2 A1 error state only	`src/ArcadeApp.tsx` rival panel, frame D22
A16	Undo for the AI strategy. After "Ask for another strategy", the player can go back to the previous one. While a new one loads, fails or is set aside, "Keep the previous strategy" restores it.	The brief asks users to be able to edit, reject, regenerate or undo AI results. Regenerating used to discard a strategy the player might have preferred.	Phase 2 "ask again" only	`src/ArcadeApp.tsx`, frames D24, D25
A17	Design tokens: one file defines the palette (background, surface, text, muted text, one accent, an AI colour, error, success), four font families, five text sizes and one spacing scale. The older stylesheets now read them.	Style basics written down once and reused on every screen and in code, as the brief requires.	Colours spread across several stylesheets	`src/styles/tokens.css`, `docs/UI_DESIGN.md`

Part B. Changes from usability testing

Filled in after the sessions, one row per issue in the Issues sheet with severity 2 or higher.

ID	Issue (severity, participants)	Change made	Status	Before / after frames
B1	*pending test data*			

5. AI tool use statement

AI tools used to build and document Farshore

Tool	Used for	Phase 3 period
Claude Code (Anthropic, Claude Opus models)	Writing and refactoring game code (3D cave and hall, AI inspector, AI Lab, commentary and voices, accounts, Google sign-in), writing automated tests, and, for Phase 3, the usability test kit, the data workbook and its formulas, automated screen captures, drafts of this document's text, and the slide layout.	October to November 2026
OpenAI Codex	Earlier development sessions: connecting the local model, the arcade engines, and the scripts that built the Phase 2 packages.	September to October 2026 [confirm]
OpenAI image generation (built-in image tool)	Original artwork: title scene, location and portrait atlases, the illustrated FARSHORE assets and trial item sprites. Prompts are recorded in `docs/ART-CREDITS.md` and `docs/FARSHORE-VISUAL-DESIGN.md`.	September to October 2026
AudioSet classifier (AST) and Whisper	Development-only listening checks that each generated sound and voice is heard as intended. Not part of the game.	October 2026

What I did myself [confirm]

- Chose the concept (an arcade journey through Homer's *Odyssey*), the eight trials and the direction for the AI rival.
- Set the requirements given to the coding assistants, reviewed their output, played every change, and decided what to keep, change or remove.
- Planned and ran every usability session in person, recorded all task outcomes, times and questionnaire answers, and wrote down participants' quotes.
- Decided which issues to fix after testing, and checked each fix in the game.
- Presented the work.

What AI tools did not do

- No participant, task result, time, SUS answer, quote or finding was generated by an AI tool. Every number in the test report is calculated from the session records in `Farshore-Phase-3-Test-Data.xlsx`.
- AI tools did not make the final design decisions. Suggestions were accepted only after review.

How AI output was checked

- Code: the automated test suite (`npm test`), the TypeScript build (`npm run build`), and manual play of each affected screen.
- Documents: read and edited by me against the game and the raw data. Calculations in the report were cross-checked against the workbook formulas.

AI inside the product (part of the design, not tool use)

- **Qwen3-4B-Instruct-2507** (`qwen3:4b-instruct`, local Ollama, or OpenAI/Groq/OpenRouter for signed-in players): chooses each rival's strategy and opening taunt before a round, and writes live bard and rival commentary. The game code owns scoring, collisions and progress.
- **Kokoro-82M**: speaks the commentary. The browser's own voice is the fallback.
- Local algorithms (A* pathfinding and scored decision rules) move the rivals during play. These are authored code, not trained models.

6. Team contribution statement

Team: Bibek Pathak (solo project).

Member	Contribution
Bibek Pathak	All Phase 3 work: game design and implementation of the tested prototype (with AI coding assistants, as described in the AI tool use statement); usability test planning, recruiting and moderating every session; data recording and analysis; the test report and list of changes; the post-test fixes; screen designs and the Figma file; slides and the presentation.

As the only member, I agree to this statement.

Signed: _____ Date: _____